L01 — study notes

Contents

Lecture 1 — Static games I: dominance and iterated elimination · Map of the notes **4**
Reading order . . . 4
Part I — representing a game (slides 2–19) . . . 4
Part II — dominance (slides 23–27) . . . 4
Part III — iterated elimination (slides 28–33) . . . 5
Support . . . 5
The one result to hold . . . 5
Numbers and cases to have memorized . . . 5
Figures . . . 5
Interactive companions . . . 6
Scope and ceiling . . . 6
Course-wide position . . . 6
Bibliography actually used . . . 7

Slides 2–14, 19 · What a game is, and how to write one down **9**
1 · U — why decision theory is not enough . . . 9
2 · C — the normal form . . . 9
2.1 · The notation for "everyone else" (slide 14; Tadelis §4.1.1, p. 59) . . . 10
2.2 · Finite two-player games are matrices (Tadelis §3.2, pp. 50–52) . . . 10
3 · Worked — the grade game (slides 10–12, 19) . . . 10
4 · Worked — the three examples of slide 4 in normal form . . . 11
4.1 · Partnership (two partners) . . . 11
4.2 · Cournot duopoly — Tadelis §3.1.2 (p. 49) . . . 11
4.3 · Second-price auction (slide 4) . . . 12
5 · G — MWG's normal form (§7.D, p. 228) . . . 12
Links . . . 12

Slides 15–18 · Static vs dynamic, complete information, and what a strategy is **13**
1 · C — static vs dynamic (slide 15) . . . 13
2 · C — complete information and common knowledge (slide 16) . . . 13
2.1 · Worked — the levels, written out . . . 13
3 · C — a strategy is a complete contingent plan (slide 17) . . . 14
3.1 · Worked — counting strategies when the pair can see your sheet . . . 14
4 · C — what a payoff is (slide 18) . . . 15
4.1 · Worked — why dominance only needs the ranking . . . 15
Links . . . 15

Slides 21–22 · What a solution concept must deliver **16**

1 · C — solution concept and equilibrium (slide 21) . . . 16
1.1 · The four maintained assumptions (Tadelis §3.3.1, pp. 54–55) . . . 16
2 · C — the three criteria (slide 22; Tadelis §3.3.2, pp. 55–56) . . . 16
3 · Worked — invariance, checked with numbers . . . 17
3.1 · Strict dominance passes (Tadelis §4.1.3, p. 62) . . . 17
3.2 · Weak dominance fails . . . 18
4 · C — the lecture's concepts on the scorecard . . . 18
Links . . . 18

Slides 23–25, 27 · Dominance and dominant-strategy equilibrium **20**
1 · C — the definitions (slide 23) . . . 20
1.1 · How many inequalities . . . 20
2 · C — dominant-strategy equilibrium (slide 25) . . . 20
3 · Worked — the prisoner's dilemma (slide 24; Tadelis p. 52) . . . 21
4 · Worked — the partnership (slide 4) . . . 21
5 · Worked — Example 3 (slide 27) and the advertising game . . . 22
6 · Worked — weak dominance in the second-price auction (slide 4) . . . 22
7 · C — weak vs strict, and a definitional trap . . . 23
7.1 · Never the best reply, yet not dominated (Cabral p. 162) . . . 24
Links . . . 24

Slide 26 · Pareto optimality, and why rational play can waste value **25**
1 · C — the definition (slide 26; Tadelis Def. 3.5, p. 57) . . . 25
2 · Worked — the prisoner's dilemma, every pair compared . . . 25
3 · Worked — the partnership's dilemma band . . . 26
4 · Worked — changing the game . . . 27
5 · C — Pareto and the other solution concepts . . . 28
Links . . . 28

Slides 27–29 · Iterated elimination of strictly dominated strategies **30**
1 · U — the idea (slide 28; Tadelis §4.2.1, pp. 63–64) . . . 30
2 · C — the procedure, formally (slide 29) . . . 30
3 · Worked — Example 3 continued: matrix (4.1), round by round . . . 31
4 · Worked — Gibbons's Figure 1.1.1 (pp. 5–7) . . . 31
5 · C — evaluating IESDS (Tadelis §4.2.3, pp. 67–68) . . . 32
6 · G — order of deletion (MWG pp. 239–240) . . . 32
Links . . . 33

Tadelis §4.2.2 · Cournot duopoly solved by iterated elimination **34**
1 · C — the game (Tadelis p. 65) . . . 34
2 · Worked — round 1: everything above 45 is strictly dominated . . . 34
3 · Worked — the general step, and round 2 . . . 35
4 · Worked — the whole sequence and its limit . . . 35
5 · C — general linear Cournot (asymmetric costs) . . . 36
6 · G — three or more firms: IESDS stops after one round . . . 37
Links . . . 37

Slides 31–33 · Spatial competition: where to open the pharmacy **38**
1 · The setup (slide 32) . . . 38
2 · Worked — (a) the normal form . . . 38

3 · Worked — (b) block 1 is strictly dominated by block 2 . . . 39
4 · Worked — (c) IESDS, round by round . . . 39
5 · Worked — (d) total walking . . . 40
6 · C — (e) what drives the prediction (Cabral §14.2) . . . 41
Links . . . 41

Lecture 1 · Formula sheet — every definition, result and number **42**
0 · The exam specification, mapped . . . 42
1 · Definitions and results . . . 43
1.1 · Model formulas . . . 44
2 · Every number in the lecture . . . 45
2.1 · Grade game (slides 10–12) . . . 45
2.2 · Prisoner's dilemma (slide 24; Tadelis p. 48) . . . 45
2.3 · Matrix (4.1) = slide 27 (Tadelis p. 64) . . . 45
2.4 · Cournot (Tadelis pp. 49, 65–68) . . . 45
2.5 · Pharmacy avenue (slides 32–33) . . . 45
2.6 · Other book examples . . . 45

Lecture 1 · Pitfalls and a True/False self-test **47**
Part I — the confusions this lecture produces . . . 47
1 · "For each s_{-i} something beats it" is not "dominated" . . . 47
2 · Strict vs weak . . . 47
3 · Equilibrium = strategies, not payoffs . . . 47
4 · Pareto optimal vs equilibrium vs fair . . . 47
5 · Mutual knowledge is not common knowledge . . . 47
6 · A strategy is not an action (once anyone observes anything) . . . 47
7 · "Dominant strategy" vs "the dominant-strategy outcome is good" . . . 47
8 · IESDS in one shot vs iteratively . . . 47
9 · IESDS reaches its limit, not a point, in Cournot . . . 48
10 · The location game's survivor needs fixed prices . . . 48
11 · "Weakly dominant equilibria need not be unique" depends on the definition . . . 48
Part II — True/False (original items) . . . 48
Justifications . . . 50
Links . . . 50

Lecture 1 · Computation in Python **51**
0 · Load the existing routines . . . 51
1 · Games as payoff matrices → L01-01-games-normal-form . . . 51
2 · Dominance, dominant-strategy equilibrium, Pareto → L01-04-dominance, L01-05-pareto-social-dilemmas . . . 52
3 · IESDS → L01-06-iesds . . . 53
4 · Partnership and changing the game → L01-05-pareto-social-dilemmas . . . 54
5 · Cournot by IESDS → L01-07-cournot-iesds . . . 54
6 · Second-price auction: weak, not strict → L01-04-dominance §6 . . . 55
7 · The pharmacy avenue → L01-08-spatial-competition . . . 55
Links . . . 56

Lecture 1 — Static games I: dominance and iterated elimination · Map of the notes

One claim. A game is a decision problem in which payoffs depend on everyone's choices, written down as $G = \langle I, \{S_i\}, \{u_i\} \rangle$. **Rationality alone removes strictly dominated strategies**, and when every player has a strictly dominant strategy the result is a unique, robust prediction — which can still be **Pareto dominated** (the prisoner's dilemma, the shirking partnership). **Common knowledge of rationality lets the removal be iterated**: IESDS solves games with no dominant strategies (matrix (4.1), Cournot, the pharmacy avenue), one level of "I know that you know" per round, and stays silent where nothing is ever dominated.

Three layers per section:

Layer	What it means here
U — intuition	Tadelis's prose (grades on a curve, the lawyer's argument, the mafia), Gibbons's §1.1, the slides' stories (grade game, pharmacies)
C — course, examinable	the slides' definitions and propositions, with Tadelis's numbering: Defs. 3.1–3.5, 4.1–4.4, Claim 4.1, Props. 4.1–4.2; the worked games
G — formal	MWG §7.D and §8.B: order of deletion, weak elimination, the three-firm Cournot failure. **Named or sketched, not examinable**

Reading order

Part I — representing a game (slides 2–19)

1. L01-01-games-normal-form — slides 2–14, 19; Tadelis pp. 43–52: the normal form and its notation across four books; the **grade game** from outcomes to payoffs; the partnership, **Cournot** (625 vs 648) and the **second-price auction** written in normal form.
2. L01-02-information-and-strategies — slides 15–18; Tadelis pp. 44–47: static means *unobserved*, not simultaneous; **common knowledge level by level**; a strategy as a contingent plan (the 2×4 grade game when Pair can see your sheet); **why dominance is ordinal**.
3. L01-03-solution-concepts — slides 21–22; Tadelis pp. 52–56: equilibrium = profile of strategies; the four maintained assumptions; **existence, uniqueness, invariance** — strict dominance passes invariance with margin $g/2$, weak dominance fails.

Part II — dominance (slides 23–27)

4. L01-04-dominance — slides 23–25, 27; Tadelis pp. 59–63: the definitions and their quantifier; **Prop. 4.1 proved**; the PD; the partnership's gain $y/n - c$; **Example 3** answered inequality by inequality; **truthful bidding is weakly, not strictly, dominant**; the weak-dominance definitional trap.
5. L01-05-pareto-social-dilemmas — slide 26; Tadelis pp. 53, 57, 62–63: Pareto sets; the **dilemma band** $y/n < c < y$; changing the game (altruism, the mafia's $z > 1$, M&R's grabbing and the investment that never happens).

Part III — iterated elimination (slides 28–33)

6. L01-06-iesds — slides 28–29; Tadelis pp. 63–68: the recursion S_i^k; matrix (4.1) **round by round with the knowledge each round uses**; **Prop. 4.2 proved**; evaluation; G: order independence (strict) and **three outcomes from three orders** (weak).
7. L01-07-cournot-iesds — Tadelis pp. 65–68: **the factorization** $(q_1 - 45)(q_1 - 45 + q_2)$, the general step, intervals $45/2^{k-1}$ converging to 30; the asymmetric closed form and exit; G: why three firms stall.
8. L01-08-spatial-competition — slides 31–33; Cabral §14.2: the mini case solved — the payoff formula, $1 \prec 2$, **(4, 4) in three rounds**, walking 12 vs 6, and why prices change the answer (Hotelling, $p = c + t$).

Support

9. L01-09-formulas — the wrap-up mapped to notes; every definition, formula and number.
10. L01-10-pitfalls — eleven confusions with counterexamples; 22 original True/False items with justifications.
11. L01-11-python — every routine with `assert` checks, **reusing** `Listas/lista-lecture-01-check.py`.

The one result to hold

strictly dominant for all $\overset{\text{Prop. 4.1}}{\Longrightarrow}$ unique DSE $\overset{\text{Prop. 4.2}}{\Longrightarrow}$ unique survivor of IESDS (one round) $\overset{\text{Lesson 2}}{\not\Longrightarrow}$ Pareto optim

Each arrow adds nothing the one before did not assume; each reverse arrow fails: IESDS can pick a unique profile with no dominant strategy anywhere (matrix (4.1)), and a unique DSE can waste value (the PD).

Numbers and cases to have memorized

Case	Numbers	Verdict
Prisoner's dilemma (slide 24)	$(-2,-2)$, $(-5,-1)$, $(-1,-5)$, $(-4,-4)$	F dominant; (F, F) Pareto dominated by (M, M)
Matrix (4.1) (slide 27)	C $\prec$ R; then M, D $\prec$ U; then R $\prec$ L	(U, L) = (4, 3) after 3 rounds
Partnership (slide 4)	gain $y/n - c$; all-E $y - c$	dilemma iff $y/n < c < y$
Cournot (Tadelis p. 65)	$[0, 45] \to [22.5, 45] \to [22.5, 33.75] \to \ldots$	$q_i = 30$, profit 900 (vs 1000 at 20)
Pharmacy avenue (slides 32–33)	removes {1,7}, {2,6}, {3,5}	(4, 4); walking 12 vs optimum 6
Mafia PD (Tadelis p. 62)	sanction z	M dominant iff $z > 1$

Figures

Built by `figures/make_figures.py` (run: `python Notas/L01/figures/make_figures.py`; ends with assert checks on the plotted numbers).

Figure	Shows	Lives in
L01-grade-game	outcome matrix vs payoff matrix; α dominant	L01-01-games-normal-form §3
L01-pd-pareto	the PD in payoff space; (F, F) inside the Pareto-improving set	L01-05-pareto-social-dilemmas §2
L01-partnership-bands	$y/n - c$ and $y - c$ against c; the dilemma band	L01-05-pareto-social-dilemmas §3
L01-spa-weak-dominance	truthful vs under/overbidding against the rival bid	L01-04-dominance §6
L01-iesds-3x3	matrix (4.1) in four rounds	L01-06-iesds §3
L01-cournot-iesds	nested squares and interval bars converging to 30	L01-07-cournot-iesds §4
L01-pharmacy-iesds	$u_A(s_A, s_B)$ heat map and the three peeling rounds	L01-08-spatial-competition §2

Interactive companions

- Shirk or Work (`companions/companion-partnership-dominance.html`) — y, c, n move the two parallel payoff lines and the cost axis with its landmarks $c = y/n$ and $c = y$; read-outs: the dominance verdict, the gain $y/n - c$, all-work vs all-shirk, the dilemma band, the waste. Defaults are the slides' PD plus 4. Linked from L01-04-dominance §4 and L01-05-pareto-social-dilemmas §3.
- The Cournot Squeeze (`companions/companion-cournot-iesds.html`) — a, b, c_1, c_2 and the round k move the best responses, the surviving rectangle and the interval bars; read-outs: S_1^k, S_2^k, the Cournot point, price, profits, rounds to 1%. Defaults are Tadelis p. 65. Linked from L01-07-cournot-iesds §5.

Both are checked by `companions/check_companions.py` (Node runs each page's model; Python recomputes it, reusing `Listas/lista-lecture-01-check.py`).

Scope and ceiling

- **Examinable:** the slides' content (wrap-up, slide 34) at the level of Tadelis Ch. 3 and §4.1–4.2: definitions with their quantifiers, Props. 4.1–4.2 with proofs, running dominance and IESDS on matrices and on Cournot/location games, Pareto comparisons.
- **Named only (G):** MWG's order-independence proof (Exercise 8.B.4), weak iterated deletion, IESDS with infinite strategy sets, three-firm Cournot.
- **Next lecture, not here:** best responses, rationalizability (Tadelis §4.3), Nash equilibrium, mixed strategies and dominance by mixtures (MWG Def. 8.B.4, Prop. 8.B.1).
- **Out of course scope:** mechanism design (the second-price auction appears only as a dominance example), cheap talk, moral hazard (`CLAUDE.md` scope block).

Course-wide position

Lecture	Relation to this one
L01	**normal form, dominance, IESDS**
Lecture 2	best responses and Nash equilibrium sharpen IESDS; mixed strategies; Hotelling voting
Lecture 3	extensive form: strategies as contingent plans (note 02 §3) pay off; backward induction
Lecture 4	multistage games and bargaining
Lecture 5	repeating the PD: cooperation the one-shot dominance argument rules out
Lecture 6	the Cournot collusion of note 07 §4 (20 vs 30) and the mafia enforcer of note 05 §4, made self-enforcing
Lecture 7	incomplete information; the second-price auction in Bayesian form
Lecture 8	signaling and perfect Bayesian equilibrium

Rules pack: rules/game-theory. **Study guide:** Map/study-guide (Lecture 1 block). **Syllabus:** rules/11_curso_programa.

Bibliography actually used

Source	Where
Slides, `Aula/Lecture01.pdf` — the spine, order and notation	every note
Tadelis (2013) Ch. 3, §4.1–4.2, pp. 43–69	the **C** layer everywhere; examples (PD, Cournot, BoS, advertising, matrix (4.1), altruistic and mafia PD)
Gibbons (1992) §1.1.A–B, pp. 2–7	normal form, PD with $T > R > P > S$, Figures 1.1.1–1.1.4 (notes 01, 04, 06)
MWG §7.D, §8.A–8.B, pp. 228, 235–241	the **G** layer: Defs. 8.B.1–8.B.3, Figure 8.B.3, order of deletion (notes 01, 04, 06, 07)
Cabral (2017) §7.1 pp. 161–163; **§14.2** pp. 354, 358–360	Figure 7.1 PD as pricing, the "for each vs one" warning, Figure 7.3; Hotelling (notes 03, 04, 05, 08)
Milgrom & Roberts (1992) Table 5.1, pp. 138–139	grabbing as a PD; hold-up (note 05)

Related project material (linked, not rebuilt):

- Practice plan: `Listas/pratica-lecture-01.md` (24 book exercises by number and page); worked solutions `Listas/pratica-lecture-01-solutions.pdf`.
- Exam-style list: `Listas/lista-lecture-01.tex` / `.pdf`, answer key `Listas/lista-lecture-01-gabarito.pdf`, and its checker `Listas/lista-lecture-01-check.py`.
- Mini-case preparation (unsolved): `Listas/prep-mini-case-lecture-01.md`.

- Quizzes: `Simulados/quiz-lecture01-1-normal-form-2026-10-13.md`, `Simulados/quiz-lecture01-2-dominance-pareto-2026-10-13.md`, `Simulados/quiz-lecture01-3-iesds-2026-10-13.md`.

Slides 2–14, 19 · What a game is, and how to write one down

Claim. A game is a decision problem in which **my payoff depends on your choice and yours on mine**, so each player must reason about the others' reasoning. The normal form $G = \langle I, \{S_i\}_{i \in I}, \{u_i\}_{i \in I} \rangle$ is the minimum bookkeeping for that: who plays, what each can do, and **how every profile of choices is valued by every player**. An outcome table (grades, prison years, prices) is not yet a game; it becomes one only when the outcomes are turned into payoffs.

Prev: — · Next: L01-02-information-and-strategies · Map: L01-00-map

---

1 · U — why decision theory is not enough

Tadelis opens Ch. 3 with the student's own problem (pp. 43–44): you choose how hard to study, but **grades are set on a curve**, so your outcome depends on how hard everyone else studies. Treating classmates as random "noise" would be inconsistent: they optimize too, and they are guessing what you will do. The slides put the same idea in three words (slide 2):

Ingredient	Meaning	New relative to Micro I?
Model	abstract from behaviour to expose the logic of strategic choice	no
Rationality	choices follow preferences, represented by a utility function (methodological individualism)	no
Interdependence	u_i depends on the actions of **all** players	**yes**

Strategic reasoning is reasoning about reasoning (slide 3): my best action depends on what I expect you to do, and what you do depends on what you expect me to do. Game theory formalizes the interaction between **preferences** and **expectations**.

Tadelis splits a **static game** into two steps (p. 44): (1) each player chooses an action *simultaneously and independently*, meaning **without observing** the others' choices (not necessarily at the same instant); (2) payoffs are distributed as a function of the profile. Gibbons makes the same point (p. 4): the prisoners need not decide at the same moment, only without knowing the other's choice.

2 · C — the normal form

Slides (13): a normal-form game is a triple $G = \langle I, \{S_i\}_{i \in I}, \{u_i\}_{i \in I} \rangle$ where

- $I = \{1, 2, \dots, n\}$ is the set of players;
- S_i is the set of (pure) strategies of player i;
- $u_i : S_1 \times \cdots \times S_n \to \mathbb{R}$ is player i's payoff function.

Tadelis, Definition 3.3 (p. 47): a finite set of players $N = \{1, \dots, n\}$, a collection of pure-strategy sets $\{S_1, \dots, S_n\}$, and payoff functions $\{v_1, \dots, v_n\}$ with $v_i : S_1 \times \cdots \times S_n \to \mathbb{R}$. **Definition 3.2 (p. 46):** a *pure strategy* is a deterministic plan of action; a profile is $s = (s_1, \dots, s_n)$ with $s_i \in S_i$.

Object	Slides	Tadelis	Gibbons (p. 4)	MWG (§7.D, §8.A)
players	I	N	$1, \dots, n$	I
payoff	u_i	v_i	u_i	u_i
the game	$\langle I, \{S_i\}, \{u_i\}\rangle$	$\langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n\rangle$	$\{S_1, \dots, S_n; u_1, \dots, u_n\}$	$\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$

These notes use the **slides' notation** (I, u_i) and cite Tadelis's numbering.

2.1 · The notation for "everyone else" (slide 14; Tadelis §4.1.1, p. 59)

$$S = S_1 \times \cdots \times S_n, \qquad s = (s_1, \dots, s_n) \in S,$$
$$s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n) \in S_{-i} = \prod_{j \neq i} S_j, \qquad s = (s_i, s_{-i}).$$

(s_i, s_{-i}) **is a reordering for reading, not a new object**: it lists the same profile with player i's entry pulled out front. Every dominance statement in this lecture is a statement "for all $s_{-i} \in S_{-i}$", so $|S_{-i}| = \prod_{j \neq i} |S_j|$ is the number of inequalities to check.

2.2 · Finite two-player games are matrices (Tadelis §3.2, pp. 50–52)

Definition 3.4 (p. 51): a *finite game* has finitely many players and finitely many strategies for each. A two-player finite game is a **bimatrix**: rows are S_1, columns are S_2, and the cell (r, c) holds $(u_1(r, c), u_2(r, c))$ — player 1's payoff first (slide 11; Gibbons p. 3). Cournot (a continuum of quantities) and the three-player voting game (pp. 49–50) are normal-form games that are **not** single matrices.

---

3 · Worked — the grade game (slides 10–12, 19)

Each student writes α or β without seeing the partner's sheet. The **outcome matrix** lists grades (You, Pair):

You \ Pair	α	β
α	B−, B−	A, C
β	C, A	B+, B+

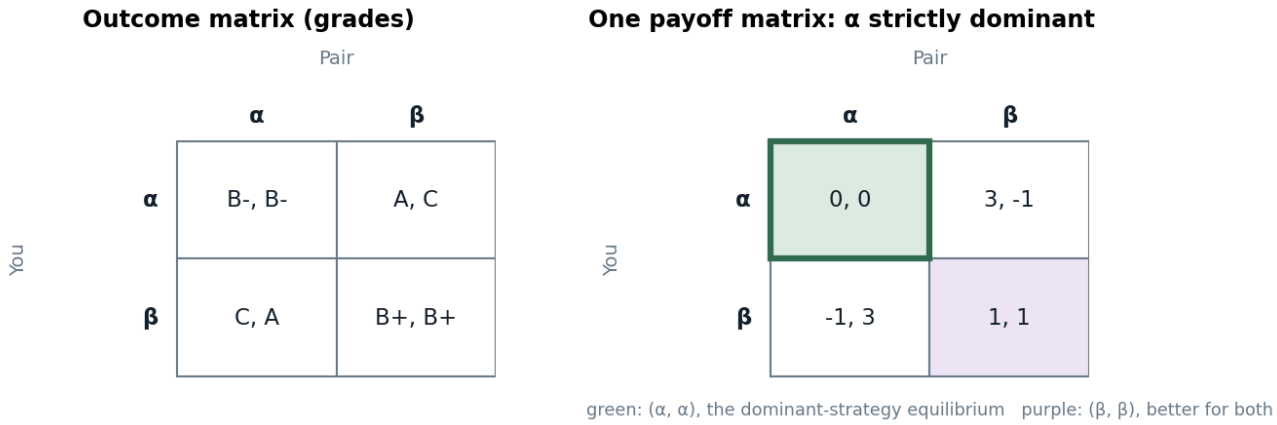
The slide's question (11) is the point: **how does each person value their own grade, and the other's?** Until that is fixed there is no game. Slide 12 proposes one set of payoffs, a selfish student with $A \mapsto 3$, B+ $\mapsto 1$, B− $\mapsto 0$, C $\mapsto -1$:

You \ Pair	α	β
α	**0, 0**	3, −1
β	−1, 3	1, 1

Example 2 (slide 19), written out. $I = \{1, 2\}$ (You, Pair); $S_1 = S_2 = \{\alpha, \beta\}$;

$$u_1(\alpha,\alpha)=0,\quad u_1(\alpha,\beta)=3,\quad u_1(\beta,\alpha)=-1,\quad u_1(\beta,\beta)=1,$$
$$u_2(s_1,s_2)=u_1(s_2,s_1)\quad\text{(the game is symmetric).}$$

Four profiles, two payoff numbers each: the eight entries of the bimatrix are the whole of $\{u_i\}$.



The same situation as an outcome table and as a game: only the right-hand table can be solved. Source: slides 11–12.

On the right, α beats β for You whatever Pair writes ($0>-1$ and $3>1$), so the selfish student writes α; by symmetry so does Pair, and both get B− although both prefer B+. That is the prisoner's dilemma of L01-04-dominance §3 and L01-05-pareto-social-dilemmas §2. Change the payoffs (a student who also values the pair's grade) and the same outcome table becomes a different game with a different prediction — Tadelis's "altruistic" prisoner's dilemma makes the point with prison years (p. 53; L01-05-pareto-social-dilemmas §4).

---

4 · Worked — the three examples of slide 4 in normal form

4.1 · Partnership (two partners)

$I=\{1,2\}$, $S_i=\{E,S\}$. Output is y times the number of partners who exert effort, split equally; effort costs the partner who exerts it $c>0$. With k efforts, each partner receives $yk/2$, so

1 \ 2	E	S
E	$y-c,\ y-c$	$\frac{y}{2}-c,\ \frac{y}{2}$
S	$\frac{y}{2},\ \frac{y}{2}-c$	$0,\ 0$

Every cell is "half of output, minus own cost": e.g. (E,S) gives output y, so partner 1 gets $y/2-c$ and partner 2 gets $y/2$. Solved in L01-04-dominance §4.

4.2 · Cournot duopoly — Tadelis §3.1.2 (p. 49)

Two identical firms, cost $c_i(q_i)=q_i^2$, inverse demand $p=100-q$ with $q=q_1+q_2$. Normal form: $N=\{1,2\}$, $S_i=[0,\infty)$, and for $i\neq j$

$$v_i(s_i, s_j) = \begin{cases} (100 - s_i - s_j)\, s_i - s_i^2 & s_i + s_j < 100 \\ -s_i^2 & s_i + s_j \geq 100 \end{cases}$$

(the second line because price cannot fall below zero). The **competitive benchmark** (p. 49, fn. 4): $MC = 2q_i = p$ gives supply $q_i = p/2$, market supply $q = p$, and $p = 100 - p$ yields $p = 50$, $q_i = 25$, profit $50 \cdot 25 - 25^2 = 625$. Cournot's objection: a firm that sees its effect on price does better. If firm 1 cuts to $q_1 = 24$, $q = 49$, $p = 51$, and

$$v_1 = 51 \cdot 24 - 24^2 = 1224 - 576 = 648 > 625.$$

So price-taking is not a rational prediction once each firm knows it moves the price; what it *should* produce depends on the rival's quantity — the strategic problem. Tadelis solves the linear-cost version by iterated elimination in §4.2.2 → L01-07-cournot-iesds.

4.3 · Second-price auction (slide 4)

$n \geq 2$ buyers, values $v_1, \ldots, v_n$ (part of the payoffs, hence common knowledge in this lecture). $S_i = [0, \infty)$ (bids). With $m_i = \max_{j \neq i} b_j$:

$$u_i(b_i, b_{-i}) = \begin{cases} v_i - m_i & b_i > m_i \\ \dfrac{v_i - m_i}{\#\{j : b_j = b_i\}} & b_i = m_i \\ 0 & b_i < m_i \end{cases}$$

The price is the **second**-highest bid, so the winner's own bid sets *whether* she wins, never *what* she pays. That separation is why bidding v_i is weakly dominant → L01-04-dominance §6.

---

5 · G — MWG's normal form (§7.D, p. 228)

MWG builds the normal form from the extensive form: a strategy is a **complete contingent plan** (a function from a player's information sets to actions), and $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ records only what each profile of plans yields. This is the "key fact" of slide 15 — every dynamic game has a normal form — and it is used from Lecture 3 on. Named here; not examinable beyond the slides' statement.

Links

- Next: information, common knowledge and strategies → L01-02-information-and-strategies.
- The games of §3–§4 solved: L01-04-dominance, L01-07-cournot-iesds.
- Python: building payoff arrays for these games → L01-11-python §1.
- Rules pack: rules/game-theory rule 1 ("specify the game completely before solving").
- Practice: `Listas/pratica-lecture-01.md` items 1–4, 9–10 (Tadelis 3.2, 3.4, 3.6, 3.7, p. 58; Muñoz-Garcia Ch. 1 Ex. 2 p. 4, Ex. 5 p. 8); quiz `Simulados/quiz-lecture01-1-normal-form-2026-10-13.md`.

Slides 15–18 · Static vs dynamic, complete information, and what a strategy is

Claim. Three words in the lecture title carry assumptions that the solution concepts will lean on. **Static** means no one observes anyone else's choice before choosing. **Complete information** means the game itself (players, actions, how outcomes map to payoffs) is **common knowledge**: everyone knows it, everyone knows everyone knows it, *ad infinitum*. And a **strategy** is a complete contingent plan, which is why even long dynamic interactions can be written as one simultaneous choice of strategies.

Prev: L01-01-games-normal-form · Next: L01-03-solution-concepts · Map: L01-00-map

---

1 · C — static vs dynamic (slide 15)

	Static (normal form)	Dynamic
What a player sees before choosing	nothing of the others' choices	some earlier moves
Defining test	choices made **without observing** the others' choices — timing is irrelevant (Tadelis p. 44; Gibbons p. 4)	someone conditions on an observed move
Natural representation	matrix / normal form	game tree (extensive form, Lecture 3)

Tadelis's example (p. 44): you study two days before the exam, I study the day before; if I do not learn how much you studied, **moving later gives me no informational advantage**, and the game is static.

Key fact (slide 15): any situation, static or dynamic, can be represented as a normal-form game by suitably defining strategies. The trick is §3 below.

2 · C — complete information and common knowledge (slide 16)

In a game of **complete information** the game being played is common knowledge: all players know who is playing, which actions each has, and how outcomes translate into payoffs (slide 16). Tadelis lists four components that must be common knowledge (p. 45): all players' possible actions, all possible outcomes, how each action profile maps into an outcome, and every player's preferences over outcomes.

Definition 3.1 (Tadelis p. 45; slide 16). An event E is *common knowledge* if (1) everyone knows E, (2) everyone knows that everyone knows E, and so on *ad infinitum*.

2.1 · Worked — the levels, written out

Let $K(E)$ be "everyone knows E". Then

$$K^1(E) = K(E), \qquad K^{m+1}(E) = K(K^m(E)), \qquad \mathrm{CK}(E) \iff K^m(E) \text{ for every } m \geq 1.$$

Situation (Tadelis p. 45)	K^1	K^2	CK?
Walking in the rain together: "it is raining"	yes	yes (each sees the other getting wet)	**yes**
Professor announces an exam; you may have been daydreaming	I know it	I am not sure **you** know it	**no** — fails at level 2

> **Mutual knowledge (K^1) is not common knowledge.** The gap matters immediately: each round of iterated elimination (L01-06-iesds) uses one more level of knowledge **of rationality**. Gibbons states the same requirement for IESDS in words (p. 7).

Common knowledge is a strong assumption (Tadelis p. 45: "by no means innocuous"); Lecture 7 drops part of it by letting payoffs be private information (Lecture 7).

3 · C — a strategy is a complete contingent plan (slide 17)

A strategy prescribes **what to do in every contingency that may arise** — the chess player who decides in advance a move for every board position (slide 17). Tadelis's candidate going to a meeting (p. 46): "if they ask q_1 I answer a_1; if they ask q_2 I answer a_2; …". MWG (§7.D): the fact that a strategy is a complete contingent plan "cannot be overemphasized".

In a static game each player has a single, unconditional decision, so **strategies coincide with actions** (Tadelis p. 47: the change of focus "may seem redundant"). The distinction pays off as soon as someone observes something.

3.1 · Worked — counting strategies when the pair can see your sheet

Take the grade game of L01-01-games-normal-form §3, but let Pair read your letter before writing (a dynamic version; original example).

- You still have one decision: $S_1 = \{\alpha, \beta\}$.
- Pair has **two contingencies** (you wrote α; you wrote β) and two actions in each, so a strategy is a pair (reply to α, reply to β): $S_2 = \{\alpha\alpha,\ \alpha\beta,\ \beta\alpha,\ \beta\beta\}$, $|S_2| = 2^2 = 4$.

With the slide-12 payoffs, the normal form is the 2×4 matrix

You \ Pair	$\alpha\alpha$	$\alpha\beta$	$\beta\alpha$	$\beta\beta$
α	0, 0	0, 0	3, −1	3, −1
β	−1, 3	1, 1	−1, 3	1, 1

Cell by cell: against $\alpha\beta$ (copy the reply "α to α, β to β"), You's α meets α (0, 0) and You's β meets β (1, 1). **The dynamic game is now a static choice of plans** — exactly slide 15's key fact. Solving it properly needs backward induction (Lecture 3).

> **Count, then draw.** With m contingencies and a actions in each, a player has a^m pure strategies. Forgetting a contingency (writing $S_2 = \{\alpha, \beta\}$ here) is the most common modelling error with sequential moves.

4 · C — what a payoff is (slide 18)

$u_i : S \to \mathbb{R}$ is player i's utility over **outcomes (strategy profiles)**; it represents i's preferences over outcomes. Stronger conditions (the expected-utility axioms) enter only "in some lectures" — when players randomize (Lecture 2).

4.1 · Worked — why dominance only needs the ranking

Let $f : \mathbb{R} \to \mathbb{R}$ be strictly increasing and replace u_i by $f \circ u_i$. Then for any s_i, s_i', s_{-i}:

$$u_i(s_i', s_{-i}) > u_i(s_i, s_{-i}) \iff f(u_i(s_i', s_{-i})) > f(u_i(s_i, s_{-i})).$$

Every comparison in the definitions of L01-04-dominance is of this form (one player, two of her strategies, a fixed s_{-i}), so **strict and weak dominance, dominant-strategy equilibrium and IESDS are unchanged by any strictly increasing transformation of each player's payoffs**. That is why the grade game only needs the *order* A $\succ$ B+ $\succ$ B− $\succ$ C (plus how much you care about the pair) — and why the numbers 3, 1, 0, −1 on slide 12 are one choice among many with the same predictions. Cardinal payoffs matter only once expectations over mixed strategies appear.

Links

- Prev / next: L01-01-games-normal-form · L01-03-solution-concepts.
- Where common knowledge of rationality does its work: L01-06-iesds §2.
- Rules pack: rules/game-theory rule 5 (common vs mutual knowledge).
- Practice: `Listas/pratica-lecture-01.md` items 3–4 (Muñoz-Garcia Ch. 1 Ex. 2 p. 4, Ex. 5 p. 8: trees to matrices); quiz `Simulados/quiz-lecture01-1-normal-form-2026-10-13.md` (sections `strategy`, `ck`).

Slides 21–22 · What a solution concept must deliver

Claim. A solution concept is a rule that maps every game into a set of profiles, its **equilibria** — the theory's prediction. It is judged on three criteria that pull against each other: **existence** (does it say something in most games?), **uniqueness** (does it say something sharp?), and **invariance** (does the answer survive a tiny change in payoffs?). "Anything can happen" always exists and is useless; "everyone plays a dominant strategy" is sharp and robust but rarely exists. Every concept in the course is a different point on that frontier.

Prev: L01-02-information-and-strategies · Next: L01-04-dominance · Map: L01-00-map

---

1 · C — solution concept and equilibrium (slide 21)

- A **solution concept** is a method for analysing games that singles out their most reasonable outcomes (slide 21; Tadelis p. 54: it restricts "the set of all possible outcomes to those that are more reasonable than others").
- An **equilibrium** is a strategy profile selected by a solution concept: the prediction.
- To be useful, a theory must apply to a wide range of strategic situations.

Caveat (Tadelis p. 61): an equilibrium is a profile of strategies, never a profile of payoffs. "(F, F) is the equilibrium; $(-4, -4)$ are the equilibrium payoffs."

1.1 · The four maintained assumptions (Tadelis §3.3.1, pp. 54–55)

#	Assumption	Content	First used by
1	**Rational**	each player chooses s_i to maximize payoff given his beliefs	strict dominance (L01-04-dominance)
2	**Intelligent**	each player knows the whole game	strict dominance (to see that a strategy is dominated)
3	**Common knowledge**	rationality and intelligence are common knowledge	IESDS (L01-06-iesds)
4	**Self-enforcement**	a prediction must be one each player is happy to follow on his own	every concept (noncooperative game theory)

Self-enforcement is why a third-party enforcer (a mafia code, a contract) has to be **modelled as part of the game**, not assumed (Tadelis p. 55; L01-05-pareto-social-dilemmas §4).

2 · C — the three criteria (slide 22; Tadelis §3.3.2, pp. 55–56)

Criterion	Question	A concept that passes trivially but is useless	A concept that is sharp but often fails
Existence	how often does it apply?	"anything can happen"	"players choose what they think the opponent chooses" — undefined when action sets differ (p. 55)
Uniqueness	how much does it restrict behaviour?	—	strict dominance: unique when it exists (Prop. 4.1) but often does not
Invariance	is it robust to payoff changes of size at most ε?	—	weak dominance (§3.2 below)

Why uniqueness is hard (slide 22; p. 56): each player's best action depends on the others' actions, so several combinations of strategies can support each other. The Battle of the Sexes (p. 54) is the canonical case:

Alex \ Chris	O	F
O	2, 1	0, 0
F	0, 0	1, 2

No strategy is "always best" for either player (O against O, F against F), so the always-best rule offers no guidance at all — a failure of **existence** for strict dominance (Tadelis p. 62), and IESDS can only say "anything can happen" here (p. 68).

What counts as a small change (p. 56): adding a player or a strategy is not small; only the payoff functions may move. Formally, if every payoff of every player moves by at most a small $\varepsilon > 0$, the prediction should not change.

3 · Worked — invariance, checked with numbers

3.1 · Strict dominance passes (Tadelis §4.1.3, p. 62)

In the prisoner's dilemma (Tadelis p. 48; L01-04-dominance §3) F strictly dominates M for player 1 with the margins

$$v_1(F, M) - v_1(M, M) = -1 - (-2) = 1, \qquad v_1(F, F) - v_1(M, F) = -4 - (-5) = 1.$$

Perturb every payoff by some δ with $|\delta| \le \varepsilon$. Each margin is a difference of two perturbed payoffs, so it moves by at most 2ε:

$$\tilde{v}_1(F, \cdot) - \tilde{v}_1(M, \cdot) \;\ge\; 1 - 2\varepsilon \;>\; 0 \quad \text{whenever } \varepsilon < \tfrac{1}{2}.$$

The same holds for player 2, so (F, F) remains the strict dominant-strategy equilibrium. Tadelis's own perturbation (5 years becoming 5.01 for one player and 4.99 for the other, p. 56) has $\varepsilon = 0.01$. In

general: **if $g > 0$ is the smallest strict margin, any perturbation with $\varepsilon < g/2$ keeps every strict inequality** — the argument behind Tadelis's sentence "because the inequality is strict, we can find a small enough value $\varepsilon > 0$" (p. 62).

3.2 · Weak dominance fails

Player 1's payoffs (player 2's are irrelevant here):

	L	R
U	1	1
D	1	0

U weakly dominates D ($1 \geq 1$, $1 > 0$). Raise $u_1(D, L)$ by any $\varepsilon/2 > 0$: now $u_1(D, L) = 1 + \varepsilon/2 > 1 = u_1(U, L)$, so U no longer weakly dominates D, **however small ε is**. A tie is not robust; a strict inequality is. (This is the counterexample used in `Listas/lista-lecture-01.tex` Q1(c) and checked in `Listas/lista-lecture-01-check.py`.)

4 · C — the lecture's concepts on the scorecard

Concept	Assumptions	Existence	Uniqueness	Invariance	Pareto optimal?
Strict dominant-strategy equilibrium (L01-04-dominance)	rationality (1–2)	rare (fails in BoS)	**yes**, Prop. 4.1	**yes**	not necessarily — PD
IESDS (L01-06-iesds)	+ common knowledge of rationality (3)	**always** in finite games	often fails (BoS: nothing removed)	yes (strict inequalities)	not necessarily — (4.1), Cournot
Nash equilibrium (Lecture 2)	+ correct beliefs	pure: not always; mixed: always (finite games)	often fails	—	not necessarily

> **Robustness to the assumptions themselves** (Cabral p. 163, Figure 7.3): when a player's best choice hinges on the *other* player being rational, a tiny doubt about that rationality can overturn it if the downside is large (Cabral's −100). The more rounds of "I know that you know" a prediction uses, the more fragile it is in practice.

Links

- Prev / next: L01-02-information-and-strategies · L01-04-dominance.
- The criteria applied: strict dominance → L01-04-dominance §5; IESDS → L01-06-iesds §5.
- Rules pack: rules/game-theory rules 2–3 (name the concept; check existence and multiplicity).

- Practice: Tadelis 4.1, 4.2 (p. 76) = `Listas/pratica-lecture-01.md` items 11–12; `Listas/lista-lecture-01.tex` Q1(c); quiz `Simulados/quiz-lecture01-1-normal-form-2026-10-13.md` (section `nf`).

Slides 23–25, 27 · Dominance and dominant-strategy equilibrium

Claim. If one strategy pays more than another **against every possible choice of the others**, a rational player never uses the worse one (Lesson 1). When every player has a strategy that beats all his others in this way, the profile of those strategies is the **strict dominant-strategy equilibrium**: it needs only rationality, it is unique (Prop. 4.1), and it is robust. Its weakness is existence: most games have no dominant strategies. Weak dominance (ties allowed) is useful — truthful bidding in a second-price auction — but it is neither robust nor a consequence of rationality alone.

Prev: L01-03-solution-concepts · Next: L01-05-pareto-social-dilemmas · Map: L01-00-map

---

1 · C — the definitions (slide 23)

Let $s_i, s_i' \in S_i$.

Term	Definition	Source
s_i **strictly dominated** by s_i'	$u_i(s_i, s_{-i}) < u_i(s_i', s_{-i})$ for **all** $s_{-i} \in S_{-i}$	slide 23; Tadelis Def. 4.1 (p. 60), written $s_i' \succ_i s_i$; MWG Def. 8.B.2
s_i **weakly dominated** by s_i'	$u_i(s_i, s_{-i}) \leq u_i(s_i', s_{-i})$ for all s_{-i}, **with strict inequality for at least one** s_{-i}	slide 23; MWG Def. 8.B.3 (p. 238)
strictly (weakly) dominant	strictly (weakly) dominates **every other** strategy of that player	slide 23; Tadelis Def. 4.2 (p. 61); MWG Def. 8.B.1

Claim 4.1 (Tadelis p. 60). *A rational player will never play a strictly dominated strategy.* If he did, the dominating strategy would give him more whatever the others do, so he could not be maximizing under any belief. This is **Lesson 1** (slide 25).

> **The quantifier is the definition.** "Strictly dominated" is "for all s_{-i}, there is one s_i' that does better" — **one fixed alternative, against everything**. It is *not* "for each s_{-i} some alternative does better" (Cabral p. 162 warns about exactly this; §7.1 below).

1.1 · How many inequalities

To show s_i' strictly dominates s_i: $|S_{-i}|$ strict inequalities. To show s_i^* strictly dominant: $(|S_i| - 1)\,|S_{-i}|$ of them. In a 3×3 game, a dominance claim between two rows is 3 inequalities; a dominant-row claim is 6.

2 · C — dominant-strategy equilibrium (slide 25)

Definition 4.3 (Tadelis p. 61; slide 25). $s^D \in S$ is a *strict dominant-strategy equilibrium* if, for every $i \in I$, s_i^D is a strictly dominant strategy.

Proposition 4.1 (Tadelis p. 61; slide 25). *If G has a strict dominant-strategy equilibrium s^D, then s^D is its unique dominant-strategy equilibrium.*

Proof. Suppose $s' \neq s^D$ were also a strict dominant-strategy equilibrium. Then $s'_i \neq s^D_i$ for some player i. Because s^D_i is strictly dominant and $s'_i \neq s^D_i$,

$$u_i(s^D_i, s_{-i}) > u_i(s'_i, s_{-i}) \quad \text{for all } s_{-i}.$$

But s'_i strictly dominant would require $u_i(s'_i, s_{-i}) > u_i(s^D_i, s_{-i})$ for all s_{-i} — the opposite inequality at every s_{-i}. Contradiction; so $s' = s^D$. ■

(Tadelis leaves this proof as exercise 4.1, p. 76.)

3 · Worked — the prisoner's dilemma (slide 24; Tadelis p. 52)

1 2	M	F
M	$-2, -2$	$-5, -1$
F	$-1, -5$	**$-4, -4$**

Player 1, F against M, column by column (the lawyer's argument, Tadelis p. 53):

Player 2 plays	$u_1(F, \cdot)$	$u_1(M, \cdot)$	F better?
M	-1	-2	$-1 > -2$ ✓
F	-4	-5	$-4 > -5$ ✓

So F strictly dominates M; with two strategies, F is strictly dominant. By symmetry the same holds for player 2, and **(F, F) is the unique strict dominant-strategy equilibrium**, with payoffs $(-4, -4)$. Gibbons's version (p. 3) uses months in jail $(-1, -9, 0, -6)$; what makes it a PD is only the ordering $T > R > P > S$ (temptation, reward, punishment, sucker; Gibbons pp. 4–5), here $-1 > -2 > -4 > -5$. The **grade game** of slide 12 has the same structure $(3 > 1 > 0 > -1)$: α is strictly dominant for both (L01-01-games-normal-form §3).

4 · Worked — the partnership (slide 4)

From the matrix of L01-01-games-normal-form §4.1, compare E with S for partner 1:

$$u_1(E, E) - u_1(S, E) = (y - c) - \tfrac{y}{2} = \tfrac{y}{2} - c, \qquad u_1(E, S) - u_1(S, S) = (\tfrac{y}{2} - c) - 0 = \tfrac{y}{2} - c.$$

The gain from effort does not depend on what the other partner does. Hence

Cost	Result
$c < y/2$	E strictly dominant for both; (E, E) the strict DSE
$c > y/2$	S strictly dominant for both; (S, S) the strict DSE
$c = y/2$	E and S pay the same against everything: neither dominates the other, **not even weakly**

With n partners and output y per effort split n ways, the gain is $y/n - c$, for the same reason: my effort raises *my* share by y/n whatever the others do. Whether the dominant choice is *good for the partnership* is a separate question → L01-05-pareto-social-dilemmas §3.

Interactive companion. Shirk or Work (`companions/companion-partnership-dominance.html`) — move c, y and n and watch the two payoff lines stay parallel: the gap $y/n - c$ is the whole dominance verdict.

5 · Worked — Example 3 (slide 27) and the advertising game

Slide 27 = Tadelis matrix (4.1), p. 64. *Does any player have a dominant strategy? Are there dominated strategies?*

1 2	L	C	R
U	4, 3	5, 1	6, 2
M	2, 1	8, 4	3, 6
D	3, 0	9, 6	2, 8

Player 1 (compare rows column by column):

Pair	L	C	R	Verdict
U vs M	$4 > 2$	$5 < 8$	$6 > 3$	no dominance
U vs D	$4 > 3$	$5 < 9$	$6 > 2$	no dominance
M vs D	$2 < 3$	$8 < 9$	$3 > 2$	no dominance

No row dominates another, so player 1 has **no dominated and no dominant strategy**.

Player 2 (compare columns row by row):

Pair	U	M	D	Verdict
R vs C	$2 > 1$	$6 > 4$	$8 > 6$	**C strictly dominated by R**
L vs R	$3 > 2$	$1 < 6$	$0 < 8$	no dominance
L vs C	$3 > 1$	$1 < 4$	$0 < 6$	no dominance

Player 2 has one dominated strategy (C) and no dominant one (L and R each win somewhere). So **no dominant-strategy equilibrium exists** — yet removing C is where IESDS starts (L01-06-iesds §3).

Tadelis's advertising game (p. 60), rows/columns L, M, H: L is strictly dominated by M for each player ($8 > 6$, $4 > 2$, $1 > 0$), but neither M nor H is dominated (M is best against M, H against H). Rationality alone removes L and stops there.

6 · Worked — weak dominance in the second-price auction (slide 4)

Bidder i with value v compares bidding v with any other bid b, for every highest rival bid m (payoffs from L01-01-games-normal-form §4.3; $k \geq 2$ is the number tied).

Underbid, $b < v$:

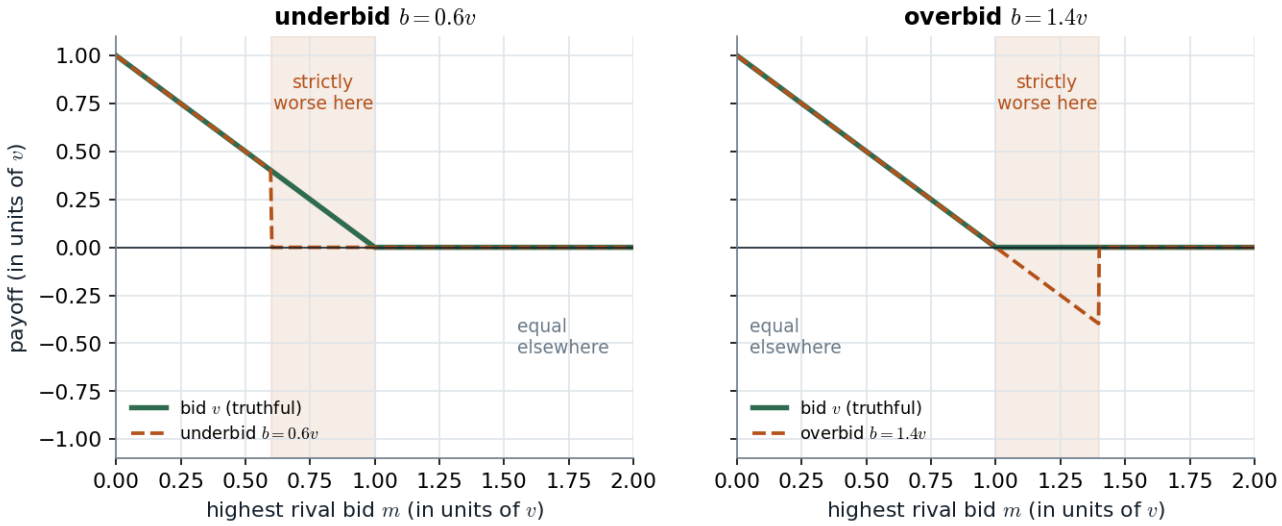
Rival bid m	Bid v	Bid b	Comparison
$m < b$	win, $v - m$	win, $v - m$	equal
$m = b$	win, $v - m > 0$	tie, $(v - m)/k$	v **strictly better**

Rival bid m	Bid v	Bid b	Comparison
$b < m < v$	win, $v - m > 0$	lose, 0	v **strictly better**
$m = v$	tie, 0	lose, 0	equal
$m > v$	lose, 0	lose, 0	equal

Overbid, $b > v$:

Rival bid m	Bid v	Bid b	Comparison
$m < v$	win, $v - m$	win, $v - m$	equal
$m = v$	tie, 0	win, $v - m = 0$	equal
$v < m < b$	lose, 0	win, $v - m < 0$	v **strictly better**
$m = b$	lose, 0	tie, $(v - m)/k < 0$	v **strictly better**
$m > b$	lose, 0	lose, 0	equal

So bidding v is **never worse and sometimes strictly better**: it weakly dominates every other bid, i.e. it is **weakly dominant**. It is *not* strictly dominant, and no bid is strictly dominated: against an m below both bids, all bids tie.



Truthful bidding loses nothing anywhere and gains in the band between the two bids: weak, not strict, dominance. Payoffs from the second-price rule of slide 4; bids in units of v (illustrative).

The shaded band is the only place the bid matters: there the underbidder loses an auction worth winning, and the overbidder wins one worth losing (paying $m > v$). Everywhere else the price is set by someone else's bid, which is the whole logic of the second-price rule.

7 · C — weak vs strict, and a definitional trap

	Strict	Weak
Follows from rationality alone?	**yes** (Claim 4.1)	no: a player sure that s_{-i} is one where the two strategies tie can rationally use the weakly dominated one (MWG p. 238)
Invariant to small payoff changes?	**yes**	**no** (L01-03-solution-concepts §3.2)
Iterated elimination order-independent?	**yes**	**no** (L01-06-iesds §6)
Dominant-strategy equilibrium unique?	**yes** (Prop. 4.1)	depends on the definition — below

> **Trap.** Tadelis's remark (p. 63) defines "weakly dominated" with $\leq$ for all s_{-i} and **no** strictness requirement, and says a weakly dominant equilibrium "need not be unique" (his exercise 4.2). With the **slides' definition** (MWG Def. 8.B.3: strict for at least one s_{-i}), a player has **at most one** weakly dominant strategy: if s and s' both weakly dominated each other, then $u_i(s,\cdot) \geq u_i(s',\cdot)$ and $u_i(s',\cdot) \geq u_i(s,\cdot)$ everywhere, so they are equal everywhere and neither can be strictly better anywhere. On an exam, state which definition you use.

7.1 · Never the best reply, yet not dominated (Cabral p. 162)

Player 1's payoffs (original numbers, in the spirit of Cabral's modified Figure 7.2):

	L	R
T	3	0
M	1	1
B	0	3

Against L, T beats M ($3 > 1$); against R, B beats M ($3 > 1$). So **for each column some row beats M** — but no single row does it against both: T fails at R ($0 < 1$), B fails at L ($0 < 1$). M is **not strictly dominated by any pure strategy**. (It is dominated by the 50–50 mix of T and B, which pays $1.5 > 1$ in both columns — dominance by mixed strategies, Lecture 2.)

Links

- Prev / next: L01-03-solution-concepts · L01-05-pareto-social-dilemmas.
- Dominated strategies removed repeatedly: L01-06-iesds. Strictly dominated strategies are never best responses (Tadelis Prop. 4.3, p. 70) and dominance by *mixed* strategies (MWG Prop. 8.B.1, p. 241) → Lecture 2.
- Python: `strictly_dominated`, `dominant`, the SPA check → L01-11-python §2, §6.
- Rules pack: rules/game-theory rule 6 ("start with dominance").
- Practice: `Listas/pratica-lecture-01.md` items 5, 11–13, 22–24 (Muñoz-Garcia Ch. 1 Ex. 13 p. 26; Tadelis 4.1, 4.2 p. 76, 4.4 pp. 76–77; MWG 8.B.1, 8.B.3, 8.B.7 p. 262); `Listas/lista-lecture-01.tex` Q1(b)–(c); quiz `Simulados/quiz-lecture01-2-dominance-pareto-2026-10-13.md` (sections `dom`, `partnership`, `auction`).

Slide 26 · Pareto optimality, and why rational play can waste value

Claim. Pareto optimality judges *outcomes* (can someone gain without anyone losing?); dominance and equilibrium predict *behaviour* (what will self-interested players do?). The two can disagree: in the prisoner's dilemma every player has a strictly dominant strategy and the resulting profile is Pareto dominated (**Lesson 2**). The disagreement has a precise source — each player ignores the effect of his choice on the others — and a precise remedy: change the game (a contract, a sanction, ownership), because no amount of reasoning inside the game removes it.

Prev: L01-04-dominance · Next: L01-06-iesds · Map: L01-00-map

---

1 · C — the definition (slide 26; Tadelis Def. 3.5, p. 57)

$s \in S$ **Pareto dominates** $s' \in S$ if $u_i(s) \geq u_i(s')$ for all $i \in I$ and $u_i(s) > u_i(s')$ for at least one i. A profile is **Pareto optimal** if no other profile Pareto dominates it.

> **Lesson 2 (slide 26): individually rational choices can lead to socially undesirable outcomes.**

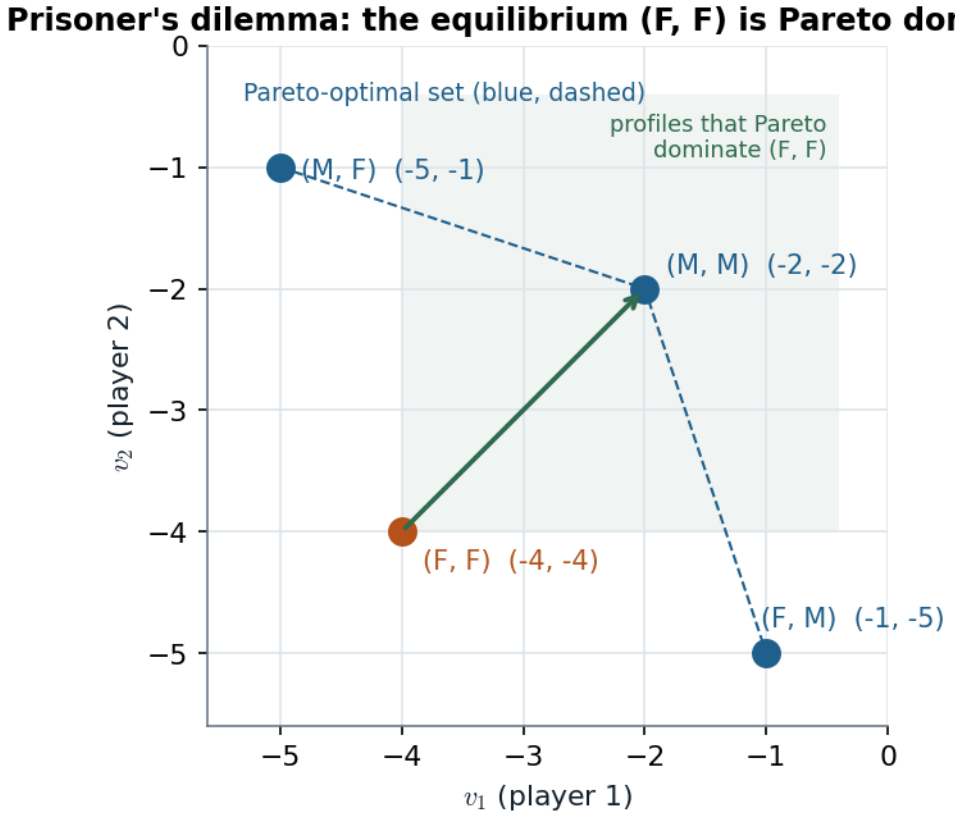
Two clarifications from Tadelis (p. 57):

- **Pareto optimal is not "fair" or "symmetric".** In the PD, (M, F) and (F, M) are Pareto optimal too: no other profile makes the sucker better off without hurting the finker.
- **The criterion is incomplete.** It cannot rank $(5, 5)$ against $(6, 60)$ in Tadelis's footnote 8 sense of equity; it only rules out *waste*. It says nothing about which Pareto optimal profile is "best".

2 · Worked — the prisoner's dilemma, every pair compared

Payoffs (Tadelis p. 48): $(M, M) = (-2, -2)$, $(M, F) = (-5, -1)$, $(F, M) = (-1, -5)$, $(F, F) = (-4, -4)$.

Profile	Dominated by	Why	Pareto optimal?
(F, F)	(M, M)	$-2 > -4$ for both	**no**
(M, M)	—	(M, F) and (F, M) each hurt one player ($-5 < -2$)	yes
(M, F)	—	anything better for 1 gives 2 less than -1	yes
(F, M)	—	symmetric	yes



The unique dominant-strategy equilibrium (orange) lies strictly inside the set of profiles that would make both players better off. Source: Tadelis p. 48 (payoffs) and Def. 3.5, p. 57.

The equilibrium is the **only** profile that is not Pareto optimal. The figure shows why the dilemma is structural: moving from (F, F) to (M, M) requires *both* players to switch, and each switch alone is a move to a sucker payoff.

The grade game (slide 12) is the same: $(\alpha, \alpha) = (0, 0)$ is the strict DSE and is Pareto dominated by $(\beta, \beta) = (1, 1)$. **Cabral's Figure 7.1** (p. 161) reads it as pricing: both firms prefer (high, high) $= (5, 5)$ to (low, low) $= (4, 4)$, yet low is dominant ($6 > 5$, $4 > 3$).

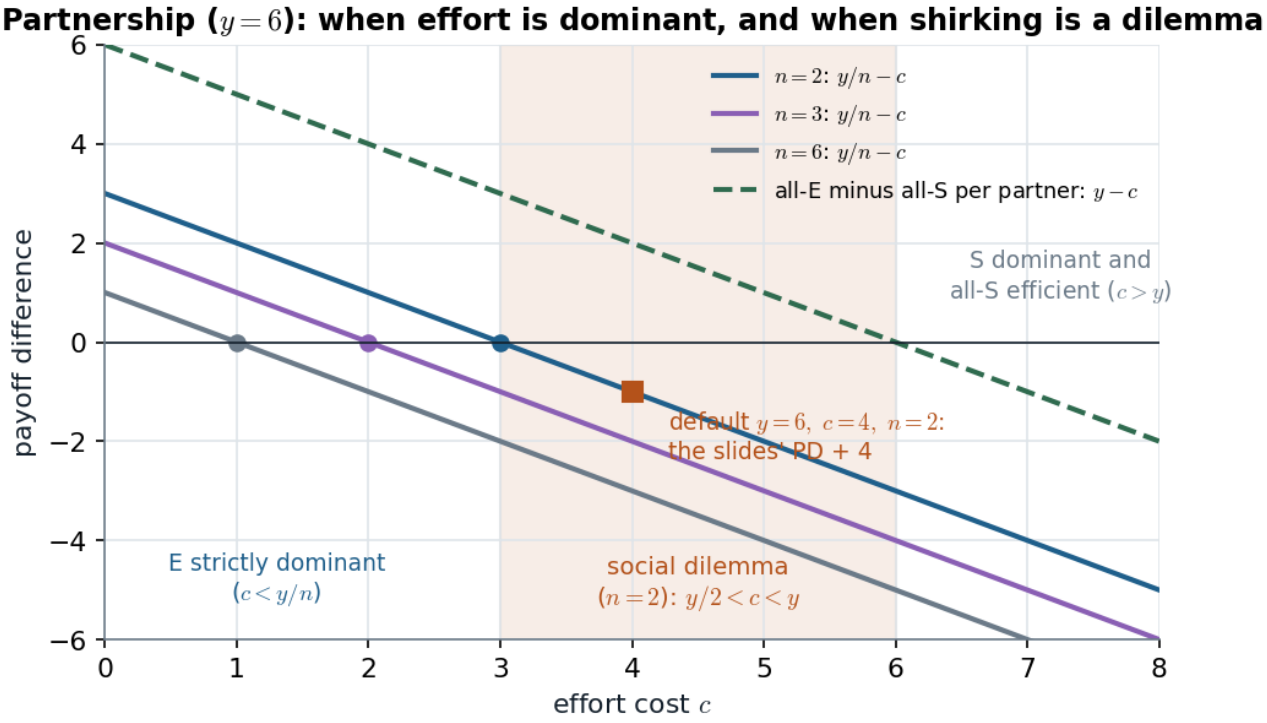
3 · Worked — the partnership's dilemma band

With n partners (slide 4, generalized), L01-04-dominance §4 gave the private gain from effort $y/n - c$, the same against every profile of the others. Compare the two symmetric profiles: all exert effort, each gets $\frac{ny}{n} - c = y - c$; all shirk, each gets 0. So:

Condition	Meaning
$c > y/n$	S strictly dominant: all-S is the strict DSE
$c < y$	all-E gives each partner $y - c > 0$: **all-E Pareto dominates all-S**
$\boxed{y/n < c < y}$	**social dilemma**: the dominant-strategy outcome is Pareto dominated

Why the band exists. Effort produces y at cost c; the partnership gains when $y > c$. But the partner who bears the whole cost receives only y/n of the product; the other $(n-1)y/n$ goes to partners who

did nothing for it. The private return understates the social return by the factor n, and the band $y/n < c < y$ is exactly where that gap changes the sign of the decision. **It widens with** n — the free-rider problem of team production.



Each solid line is the private gain from effort, $y/n-c$; the dashed line is the per-partner gain from everyone working, $y-c$. Between the two zeros, shirking is dominant and wasteful. Source: slide 4's partnership with $y=6$; the square ($c=4$, $n=2$) is Tadelis's PD (p. 48) with 4 added to every payoff.

At $y=6$, $c=4$, $n=2$ the matrix is $(E,E)=(2,2)$, $(E,S)=(-1,3)$, $(S,E)=(3,-1)$, $(S,S)=(0,0)$ — the slides' prisoner's dilemma shifted by $+4$, so the partnership **is** a PD whenever it is in the band. Adding partners at fixed y moves the zero of the private line left (y/n) while the social zero stays at y: the band only grows.

> **Interactive companion.** Shirk or Work (`companions/companion-partnership-dominance.html`) — set n and drag c across the two landmarks $c=y/n$ and $c=y$; the verdict switches from "effort dominant" to "dilemma" to "shirking dominant and efficient".

4 · Worked — changing the game

The altruistic PD (Tadelis p. 53). If each year in prison costs the prisoner 1 and his partner $\frac{1}{2}$:

1 2	M	F
M	$-2-1=-3,\ -3$	$-5-\frac{1}{2}=-5\frac{1}{2},\ -1-\frac{5}{2}=-3\frac{1}{2}$
F	$-3\frac{1}{2},\ -5\frac{1}{2}$	$-4-2=-6,\ -6$

Now M is strictly dominant ($-3 > -3\frac{1}{2},\ -5\frac{1}{2} > -6$): cooperation, because the *payoffs* changed. "Garbage in, garbage out" (p. 53).

The mafia-modified PD (Tadelis pp. 62–63). A sanction z on every finker:

1 2	M	F
M	$-2, -2$	$-5, -1-\mathrm{z}$
F	$-1-\mathrm{z}, -5$	$-4-\mathrm{z}, -4-\mathrm{z}$

M beats F for player 1 against M iff $-2 > -1-z \iff z > 1$, and against F iff $-5 > -4-z \iff z > 1$. So **any sanction $z > 1$ makes M strictly dominant** and (M, M) the Pareto-optimal DSE. If players believe the code is enforced, it is never used. Whether the enforcer will actually punish is a question about *his* incentives — repeated games, Lecture 5 and Lecture 6.

Milgrom & Roberts, Table 5.1 (pp. 138–139): grabbing in a joint venture. Each firm invests 2 in a cospecialized asset; gross return 8; grabbing costs 3 and takes everything if the other does not grab. Payoffs (A as row player; M&R print A on the columns):

A B	Grab	Don't
Grab	$4-2-3=-1,\ -1$	$8-2-3=3,\ 0-2=-2$
Don't	$-2,\ 3$	$4-2=2,\ 2$

Grab is strictly dominant ($3 > 2$, $-1 > -2$), so (Grab, Grab) $= (-1, -1)$ is the DSE, Pareto dominated by $(2, 2)$. M&R's twist (p. 139): **the firms do not have to play**. Anticipating -1 each, they compare investing (-1) with not investing (0) and do not invest — the hold-up problem: opportunism that cannot be contracted away destroys the investment itself. Remedies they list: common ownership, reputation, costly commitment.

5 · C — Pareto and the other solution concepts

Game	Prediction	Pareto dominated by	Source
PD	(F, F), payoffs $(-4, -4)$	(M, M)	Tadelis p. 62
Matrix (4.1)	IESDS: (U, L), payoffs (4, 3)	(M, C) = (8, 4), (D, C) = (9, 6)	Tadelis p. 68
Cournot, $p = 100 - q$, $c = 10$	IESDS: $q_i = 30$, profit 900	$q_i = 20$: profit 1000	Tadelis p. 68

> **The failure of Pareto optimality is not a failure of the solution concept** (Tadelis p. 62): it is a correct prediction that the players, left to their own devices and unable to make binding agreements, waste value. It points to where institutions pay.

Links

- Prev / next: L01-04-dominance · L01-06-iesds.
- The same tension with Nash equilibrium and repeated play: Lecture 2, Lecture 5, Lecture 6.
- Python: `pareto_optimal`, the partnership band, the mafia threshold → L01-11-python §2, §4.
- Rules pack: rules/game-theory rule 20 (zero-sum framing of positive-sum play); M&R hold-up in rules/06.

- Practice: `Listas/lista-lecture-01.tex` Q1(a) (partnership with synergy), Q1(d) (Pareto with n players), Q2 (continuous effort); quiz `Simulados/quiz-lecture01-2-dominance-pareto-2026-10-13.md` (sections `partnership`, `pareto`); `Listas/pratica-lecture-01.md` item 13 (MWG 8.B.1, p. 262).

Slides 27–29 · Iterated elimination of strictly dominated strategies

Claim. Rationality removes strictly dominated strategies; **common knowledge of rationality** lets every player remove the others' dominated strategies too, look at the smaller game, and remove what has *become* dominated there — again and again. Each round costs one more level of "I know that you know that …". The procedure always exists in finite games, its result does not depend on the order of removal, it contains any strict dominant-strategy equilibrium as its unique survivor (Prop. 4.2), and it often leaves many profiles standing.

Prev: L01-05-pareto-social-dilemmas · Next: L01-07-cournot-iesds · Map: L01-00-map

---

1 · U — the idea (slide 28; Tadelis §4.2.1, pp. 63–64)

1. A rational player never plays a strictly dominated strategy (Claim 4.1).
2. If all players know this, each can **ignore the others' strictly dominated strategies** — everyone is effectively playing a smaller game.
3. In the smaller game, strategies that were not dominated before may be dominated now; and the others can reason the same way.
4. This works because **rationality is common knowledge** (slide 28); repeat until nothing more can be removed.

2 · C — the procedure, formally (slide 29)

Let $S_i^0 = S_i$. For each $k \geq 1$ and each player i,

$$S_i^k = \Big\{ s_i \in S_i^{k-1} \,:\, \nexists\, s_i' \in S_i^{k-1} \text{ such that } u_i(s_i', s_{-i}) > u_i(s_i, s_{-i}) \;\; \forall s_{-i} \in S_{-i}^{k-1} \Big\},$$

$$S_i^\infty = \bigcap_{k \geq 1} S_i^k.$$

Read the quantifiers: the dominator s_i' must itself still be available (S_i^{k-1}), and the inequality is required only against the opponents' **surviving** profiles (S_{-i}^{k-1}) — that restriction is where knowledge of the others' rationality enters. Every player is pruned **simultaneously** in each round.

Tadelis gives the same thing as an algorithm (p. 65): set $S_i^0 = S_i$, $k = 0$; if some $s_i \in S_i^k$ is strictly dominated, remove every such strategy for every player, set $k = k + 1$ and repeat; otherwise stop — the remaining strategies are the prediction.

Definition 4.4 (Tadelis p. 65; slide 29). Any profile $s^{ES} = (s_1^{ES}, \ldots, s_n^{ES})$ that survives IESDS is an **iterated-elimination equilibrium**.

Proposition 4.2 (Tadelis p. 68; slide 29). *If s^* is a strict dominant-strategy equilibrium, then s^* is the unique profile that survives IESDS.*

Proof. For each i, s_i^* strictly dominates every other $s_i \in S_i$ against all of $S_{-i} = S_{-i}^0$. So in round 1 every $s_i \neq s_i^*$ is removed and s_i^* is not (nothing dominates it: it strictly beats everything). Hence $S_i^1 = \{s_i^*\}$ for all i; a singleton has nothing left to remove, so $S^\infty = \{s^*\}$. ■

One round needs only rationality; every later round needs one more level of knowledge. That is why Prop. 4.2's survivor is as robust as the DSE itself, while an IESDS prediction reached after many rounds is only as credible as a long chain of mutual knowledge.

3 · Worked — Example 3 continued: matrix (4.1), round by round

The game of slide 27 (Tadelis (4.1), p. 64). L01-04-dominance §5 found no dominated strategy for player 1 and one for player 2.

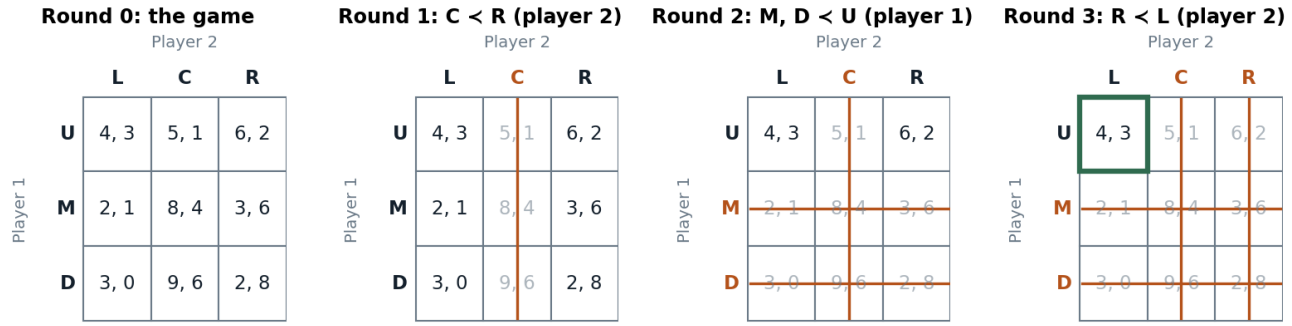
Round 1 (needs: player 2 is rational). C $\prec$ R for player 2: $2 > 1$ (U), $6 > 4$ (M), $8 > 6$ (D). Nothing for player 1. $S_1^1 = \{U, M, D\}$, $S_2^1 = \{L, R\}$.

1 2	L	R
U	4, 3	6, 2
M	2, 1	3, 6
D	3, 0	2, 8

Round 2 (needs: player 1 is rational **and knows** player 2 is rational). Against $\{L, R\}$ only: U vs M: $4 > 2$, $6 > 3$; U vs D: $4 > 3$, $6 > 2$. **M and D are both strictly dominated by U** — neither was dominated in the full game, because C (where they paid 8 and 9) was still possible. $S_1^2 = \{U\}$, $S_2^2 = \{L, R\}$.

Round 3 (needs: player 2 is rational, knows player 1 is rational, and knows that player 1 knows player 2 is rational). Against U only: $u_2(U, L) = 3 > 2 = u_2(U, R)$, so R $\prec$ L. $S_1^3 = \{U\}$, $S_2^3 = \{L\}$.

Round 4: nothing left to remove. $S^\infty = \{(U, L)\}$, payoffs $(4, 3)$ (Tadelis p. 65).



IESDS on matrix (4.1): each round strikes what the previous rounds made dominated. Source: slide 27; Tadelis p. 64–65.

Read the second panel against the third: M and D survive round 1 only because column C protects them, and they fall the moment C is gone. Elimination is **cumulative** — later rounds are possible only because earlier ones happened. Note also that the survivor (4, 3) is Pareto dominated by (M, C) and (D, C) (L01-05-pareto-social-dilemmas §5).

4 · Worked — Gibbons's Figure 1.1.1 (pp. 5–7)

1 2	Left	Middle	Right
Up	1, 0	1, 2	0, 1
Down	0, 3	0, 1	2, 0

Round	Removed	Inequalities	Knowledge used
1	Right (player 2)	Middle beats Right: $2 > 1$, $1 > 0$	2 rational
2	Down (player 1)	Up beats Down against {Left, Middle}: $1 > 0$, $1 > 0$	1 rational, knows 2 rational
3	Left (player 2)	Middle beats Left against Up: $2 > 0$	2 knows 1 knows 2 rational

Survivor: **(Up, Middle)**. In the full game Up and Down are not ranked (Up wins at Left, Down at Right) — Down falls only once Right is gone.

5 · C — evaluating IESDS (Tadelis §4.2.3, pp. 67–68)

Criterion	Verdict	Why
Existence	always, in finite games	the procedure can always be run; each S_i^k stays nonempty (a strategy that is best against some fixed s_{-i} is never strictly dominated) and it stops after at most $\sum_i(\lvert S_i\rvert - 1)$ rounds
Uniqueness	often fails	Battle of the Sexes: nothing is dominated, "anything can happen" (p. 68); Gibbons's Figure 1.1.4 (p. 7): a 3×3 game with no dominated strategy at all
Invariance	holds	every removal rests on strict inequalities (L01-03-solution-concepts §3.1)
Pareto	not guaranteed	PD, matrix (4.1), Cournot (L01-05-pareto-social-dilemmas §5)
Assumptions	heavy	common knowledge of rationality; Gibbons's first "drawback" (p. 7); Tadelis: players must be "extremely intelligent" (p. 68)

> **IESDS is weaker than dominant-strategy equilibrium in what it predicts, stronger in where it applies.** It predicts exactly the DSE when one exists (Prop. 4.2), predicts something sharp in games with no dominant strategies (matrix (4.1), Cournot), and predicts nothing in games like the Battle of the Sexes.

6 · G — order of deletion (MWG pp. 239–240)

Strict: order does not matter. "If at any given point several strategies (of one or several players) are strictly dominated, then we can eliminate them all at once or in any sequence without changing

the set of strategies that we ultimately end up with" (MWG p. 239–240; proof is MWG Exercise 8.B.4, p. 262). The reason: if s_i' beats s_i against a set of opponent profiles, it still beats it against any subset, so a strategy dominated now stays dominated after other removals — as long as a dominator survives, which in a finite game is guaranteed because a removed dominator was itself beaten by a survivor, and strict dominance is transitive.

Weak: order matters (MWG Figure 8.B.3, p. 238):

1 2	L	R
U	5, 1	4, 0
M	6, 0	3, 1
D	6, 4	4, 4

D weakly dominates U ($6 > 5$, $4 = 4$) and M ($6 = 6$, $4 > 3$).

Order	Step 1	Step 2 (player 2 on the remaining rows)	Step 3	Prediction
remove U first	rows {M, D}	R beats L: $1 > 0$, $4 = 4 \rightarrow$ remove L	at R: D $4 > 3$ M $\rightarrow$ remove M	**(D, R)**
remove M first	rows {U, D}	L beats R: $1 > 0$, $4 = 4 \rightarrow$ remove R	at L: D $6 > 5$ U $\rightarrow$ remove U	**(D, L)**

A third order — remove U and then M before touching player 2 — leaves $\{D\} \times \{L, R\}$, where L and R tie at 4 and nothing more can go. Three orders, three different predictions — and player 2's payoff differs only through which of his "ties" was broken first. MWG's diagnosis (p. 240): weak elimination assumes a player assigns positive probability to every rival strategy, but iterated deletion assumes the deleted ones are never played. The slides' IESDS uses **strict** dominance only.

Infinite strategy sets (G): with $S = [0, 1)$ and $u(s) = s$, every s is strictly dominated by $(s + 1)/2$, so nothing survives round 1 — existence needs finiteness (or compactness and continuity). Cournot (L01-07-cournot-iesds) survives because profit is continuous and strategies can be bounded.

Links

- Prev / next: L01-05-pareto-social-dilemmas · L01-07-cournot-iesds.
- The same logic on a line: L01-08-spatial-competition. IESDS survivors as best responses (Tadelis Prop. 4.4, p. 70), rationalizability and Nash → Lecture 2.
- Python: `iesds2` and the round log, reused from `Listas/lista-lecture-01-check.py` → L01-11-python §3.
- Rules pack: rules/game-theory rules 5–6.
- Practice: `Listas/pratica-lecture-01.md` items 6–8, 14, 19–20 (Tadelis 4.5 p. 77; Gibbons 1.2 p. 48; Muñoz-Garcia Ch. 1 Ex. 9 p. 17, Ex. 12 p. 23, Ex. 16 p. 31; MWG 8.B.4 p. 262); `Listas/lista-lecture-01.tex` Q3; quiz `Simulados/quiz-lecture01-3-iesds-2026-10-13.md` (sections `props`, `ck`).

Tadelis §4.2.2 · Cournot duopoly solved by iterated elimination

Claim. With linear demand and constant marginal cost, a quantity above the monopoly output is strictly dominated (round 1); once rivals are known to stay below it, quantities too *low* become dominated (round 2); and so on. The surviving interval halves every round, alternately from above and below, and shrinks to the single point $q_1 = q_2 = 30$ in Tadelis's example — a sharp prediction from dominance alone, in a game with a **continuum** of strategies and no dominant strategy. The prediction is not Pareto optimal for the firms: each would earn more if both produced less.

Prev: L01-06-iesds · Next: L01-08-spatial-competition · Map: L01-00-map

---

1 · C — the game (Tadelis p. 65)

Two firms, cost $c_i(q_i) = 10q_i$, inverse demand $p = 100 - q$, $q = q_1 + q_2$. Firm 1's payoff:

$$v_1(q_1, q_2) = \underbrace{(100 - q_1 - q_2)\, q_1}_{\text{revenue}} - \underbrace{10q_1}_{\text{cost}} = 90q_1 - q_1^2 - q_1 q_2.$$

Best response. For a given q_2, v_1 is concave in q_1; the first-order condition $90 - 2q_1 - q_2 = 0$ gives

$$q_1 = \frac{90 - q_2}{2} \qquad (4.2)$$

and symmetrically for firm 2. (If $q_2 \geq 90$ the best response is $q_1 = 0$.)

2 · Worked — round 1: everything above 45 is strictly dominated

Since $q_2 \geq 0$, (4.2) never exceeds 45. Tadelis shows the dominance directly (p. 66):

$$v_1(45, q_2) = (100 - 45 - q_2)45 - 450 = 2025 - 45q_2,$$

$$v_1(45, q_2) - v_1(q_1, q_2) = 2025 - 45q_2 - 90q_1 + q_1 q_2 + q_1^2.$$

Group the terms (a factorization the book leaves as "easy to check", fn. 3):

$$= \underbrace{(q_1^2 - 90q_1 + 2025)}_{(q_1-45)^2} + \underbrace{(q_1 q_2 - 45q_2)}_{q_2(q_1-45)} = (q_1 - 45)\,(q_1 - 45 + q_2).$$

For $q_1 > 45$ and any $q_2 \geq 0$ both factors are positive, so **every $q_1 > 45$ is strictly dominated by $q_1 = 45$.** Same for firm 2. After round 1: $q_i \in [0, 45]$.

3 · Worked — the general step, and round 2

Lemma (the step every round uses). Write $v_i(x, q_j) - v_i(z, q_j)$ for two own quantities:

$$v_i(x, q_j) - v_i(z, q_j) = 90(x - z) - (x^2 - z^2) - q_j(x - z) = (x - z)\,(90 - x - z - q_j).$$

- If the rival is known to produce **at most** U, let $x = (90 - U)/2$ (the best reply to U). For any $z < x$ and any $q_j \le U$: $90 - x - z - q_j > 90 - 2x - U = 0$, so x strictly beats z. **Quantities below $BR(U)$ are dominated.**
- If the rival is known to produce **at least** L, let $x = (90 - L)/2$. For any $z > x$ and $q_j \ge L$: $90 - x - z - q_j < 90 - 2x - L = 0$, and $x - z < 0$, so the product is again positive. **Quantities above $BR(L)$ are dominated.**

Round 2. Rivals are in $[0, 45]$, so $L = 0, U = 45$: the lower bound becomes $BR(45) = (90-45)/2 = 22.5$ and the upper bound stays $BR(0) = 45$. Surviving: $[22.5, 45]$.

Round 3. Rivals in $[22.5, 45]$: lower bound $BR(45) = 22.5$, upper bound $BR(22.5) = (90 - 22.5)/2 = 33.75$. Surviving: $[22.5,\ 33\frac{3}{4}]$ (Tadelis p. 66).

4 · Worked — the whole sequence and its limit

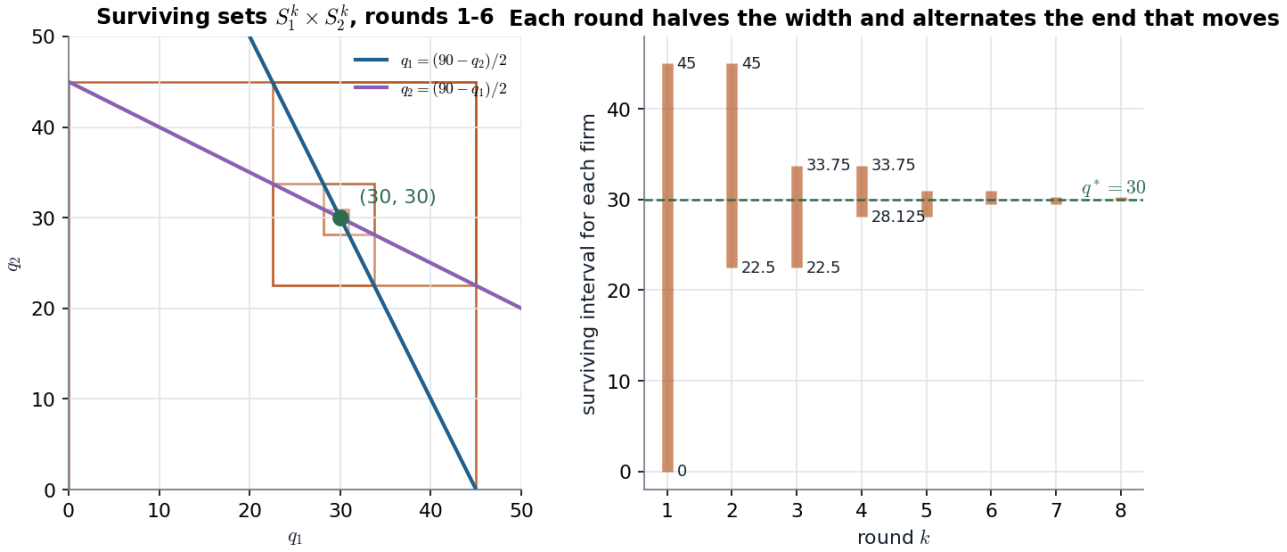
The recursion is $L^k = BR(U^{k-1})$, $U^k = BR(L^{k-1})$, starting from $L^0 = 0$ and no upper bound (BR of "anything" is at most 45).

Round k	L^k	U^k	Width $U^k - L^k$	Endpoint that moved
1	0	45	45	upper
2	22.5	45	22.5	lower
3	22.5	33.75	11.25	upper
4	28.125	33.75	5.625	lower
5	28.125	30.9375	2.8125	upper
6	29.53125	30.9375	1.40625	lower

Why it halves. Measure from 30: $BR(q) - 30 = \frac{90-q}{2} - 30 = -\frac{q-30}{2}$. Each application of the best response **halves the distance to 30 and flips its side**. The endpoint updated at round k is therefore

$$30 + 15\left(-\tfrac{1}{2}\right)^{k-1} \quad (45,\ 22.5,\ 33.75,\ 28.125, \ldots), \qquad \text{width} = \frac{45}{2^{k-1}}.$$

The limit (Tadelis p. 67). If the process stopped at an interval $[q_{\min}, q_{\max}]$, by symmetry $q_{\min} = \frac{90-q_{\max}}{2}$ and $q_{\max} = \frac{90-q_{\min}}{2}$. Subtracting, $q_{\max} - q_{\min} = \frac{q_{\max}-q_{\min}}{2}$, so $q_{\max} = q_{\min} = q$, and $q = \frac{90-q}{2}$ gives $\mathbf{q = 30}$. The width is positive after every finite round; the prediction is the intersection $S_i^\infty = \bigcap_k S_i^k = \{30\}$.



IESDS in the Cournot game: each square is the set of profiles still standing; the bars are the same intervals for one firm. Source: Tadelis eq. (4.2) and Figure 4.1, pp. 66–67; $p = 100 - q$, $c_i = 10$.

The left panel is Tadelis's Figure 4.1 drawn for both firms at once: the upper end of one round's interval sets the lower end of the next through the best-response line. The right panel makes the halving visible — after 6 rounds both firms are within 1.5 units of 30.

Profits (Tadelis p. 68). At $q_1 = q_2 = 30$: $p = 40$, $v_i = (40 - 10) \cdot 30 = 900$. If both produced 20: $p = 60$, $v_i = 50 \cdot 20 = 1000$. The joint optimum maximizes $(90 - q)q$ at $q = 45$, i.e. 22.5 each, $p = 55$, $v_i = 45 \cdot 22.5 = 1012.5$. **IESDS is not Pareto optimal for the firms** (L01-05-pareto-social-dilemmas §5); collusion and its enforcement are Lecture 6.

5 · C — general linear Cournot (asymmetric costs)

Demand $p = a - bq$, costs $c_1, c_2 < a$. Then $BR_i(q_j) = \max\{0, \frac{a - c_i - bq_j}{2b}\}$, round 1 removes $q_i > m_i = \frac{a - c_i}{2b}$ (the monopoly output), and the intervals converge to the intersection of the best responses. Solving $2bq_1 = a - c_1 - bq_2$ and $2bq_2 = a - c_2 - bq_1$: double the first and substitute the second,

$$4bq_1 = 2a - 2c_1 - (a - c_2 - bq_1) \implies 3bq_1 = a - 2c_1 + c_2,$$

$$\boxed{q_i^* = \frac{a - 2c_i + c_j}{3b}}, \qquad p^* = \frac{a + c_1 + c_2}{3}, \qquad v_i^* = \frac{(a - 2c_i + c_j)^2}{9b}.$$

Check with Tadelis's numbers: $q^* = (100 - 20 + 10)/3 = 30$, $p^* = 40$, $v^* = 90^2/9 = 900$. ✓

Change	Effect on q_1^*	Reading
$a \uparrow$	$+\frac{1}{3b}$	bigger market
$c_1 \uparrow$	$-\frac{2}{3b}$	own cost: twice the weight
$c_2 \uparrow$	$+\frac{1}{3b}$	rival weaker, I expand
$b \uparrow$	$q_i^* \propto 1/b$	steeper demand shrinks both

Exit. If $a - 2c_1 + c_2 \leq 0$, i.e. $c_1 \geq \frac{a+c_2}{2}$, firm 1's interval collapses to $\{0\}$ and firm 2 produces its monopoly output $\frac{a-c_2}{2b}$. With strict inequality firm 1 is eliminated after finitely many rounds (checked in `Listas/lista-lecture-01-check.py`, Q4).

> **Interactive companion.** The Cournot Squeeze (`companions/companion-cournot-iesds.html`) — set a, b, c_1, c_2 and step through rounds k: the nested rectangles close in on the Cournot point; raise c_1 past $(a + c_2)/2$ and watch firm 1 squeezed out.

6 · G — three or more firms: IESDS stops after one round

Gibbons warns (p. 7) that with three or more Cournot firms IESDS removes almost nothing (MWG Exercise 8.B.5, p. 262, asks for the argument). Sketch with identical firms: round 1 leaves $q_j \in [0, m]$, $m = \frac{a-c}{2b}$. Firm i now faces $Q_{-i} = \sum_{j \neq i} q_j \in [0, 2m]$, and its best reply $m - Q_{-i}/2$ sweeps **all** of $[0, m]$ as Q_{-i} ranges over $[0, 2m]$. A quantity that is the unique best reply to some surviving Q_{-i} cannot be strictly dominated, so nothing more is removed: the prediction is the cube $[0, m]^3$. With two firms the rival alone can never push Q_{-i} to $2m$, which is what makes the interval shrink. The sharper tool for n firms is Nash equilibrium → Lecture 2 (Gibbons §1.2.A).

Links

- Prev / next: L01-06-iesds · L01-08-spatial-competition.
- The normal form of Cournot (quadratic-cost version, Tadelis p. 49): L01-01-games-normal-form §4.2. Best responses and the Nash equilibrium of the same model → Lecture 2; leadership (Stackelberg) → Lecture 3.
- Python: `cournot_iesds` reused from `Listas/lista-lecture-01-check.py` → L01-11-python §5.
- Practice: `Listas/pratica-lecture-01.md` items 15, 21 (Tadelis 4.6, p. 77; MWG 8.B.5, p. 262); `Listas/lista-lecture-01.tex` Q4 (asymmetric costs); quiz `Simulados/quiz-lecture01-3-iesds-2026-10-13.md` (section `cournot`).

Slides 31–33 · Spatial competition: where to open the pharmacy

Claim. On a line of equally populated blocks, a store at the edge is strictly dominated by the block next to it, because moving inward keeps every customer it had and wins some of the rival's. Common knowledge of rationality peels the avenue from both ends, one layer per round, until **both chains sit on the middle block** — the clustering of rival pharmacies on the same street. The prediction doubles the customers' total walking relative to the best pair of locations, and it rests on fixed, equal prices: once prices are chosen too, being next to the rival invites a price war and firms spread out (Cabral §14.2).

Prev: L01-07-cournot-iesds · Next: L01-09-formulas · Map: L01-00-map

Spoiler. This note solves the in-class mini case (handed in on Oct 13). To practise first, work `Listas/prep-mini-case-lecture-01.md` (different streets, same skills) and Tadelis 4.10 (p. 78).

---

1 · The setup (slide 32)

Two chains, A and B, each open one store on a 7-block avenue; one customer per block; both choose a block simultaneously; identical prices and products; each customer buys from the nearest store and splits evenly when equidistant; each chain maximizes its number of customers. The motivation (slide 31): rival pharmacies opening right next to each other in Brasília.

2 · Worked — (a) the normal form

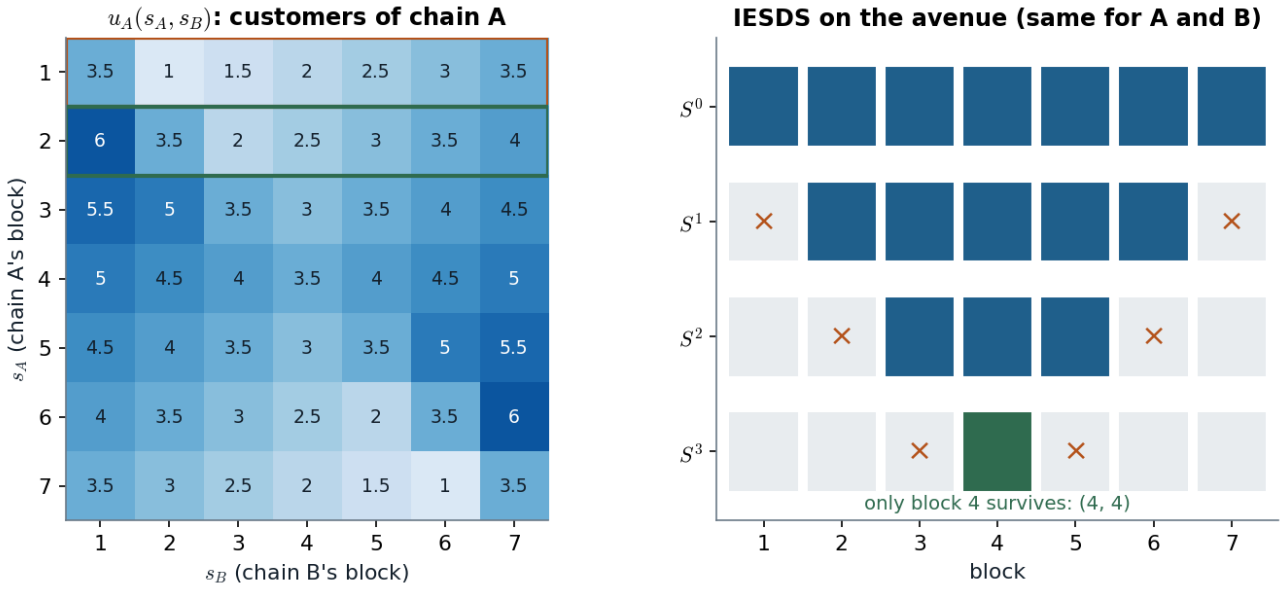
$I = \{A, B\}$, $S_A = S_B = \{1, \dots, 7\}$. Since every customer buys from someone, $u_A + u_B = 7$ in every cell (a **constant-sum** game).

Case $s_A < s_B$. Block j goes to A when $j - s_A < s_B - j$, i.e. $j < \frac{s_A+s_B}{2}$.

- Blocks $1, \dots, s_A$ are all closer to A: s_A customers.
- If $s_A + s_B$ is odd, the midpoint is not a block; blocks $s_A + 1, \dots, \frac{s_A+s_B-1}{2}$ go to A: $\frac{s_B-s_A-1}{2}$ more. Total $s_A + \frac{s_B-s_A-1}{2} = \frac{s_A+s_B-1}{2}$.
- If $s_A + s_B$ is even, block $\frac{s_A+s_B}{2}$ is equidistant and splits; A gets blocks $s_A + 1, \dots, \frac{s_A+s_B}{2} - 1$ plus half the midpoint: $s_A + (\frac{s_A+s_B}{2} - 1 - s_A) + \frac{1}{2} = \frac{s_A+s_B-1}{2}$. **Same formula.**

$$u_A(s_A, s_B) = \begin{cases} \dfrac{s_A + s_B - 1}{2} & s_A < s_B \\ \dfrac{7}{2} & s_A = s_B \\ 7 - \dfrac{s_A + s_B - 1}{2} & s_A > s_B \end{cases}$$

(the third line is $7 - u_B$ with the roles swapped; at $s_A = s_B$ every block is equidistant and splits). Example: $u_A(2, 5) = \frac{2+5-1}{2} = 3$ — blocks 1, 2, 3; block 4 is closer to 5 ($1 < 2$).



Row 2 beats row 1 cell by cell (the outlined rows); elimination removes the two outermost surviving blocks each round. Source: slides 32–33; payoffs from the formula above.

3 · Worked — (b) block 1 is strictly dominated by block 2

Compare the two outlined rows of the heat map, against **every** s_B:

s_B	1	2	3	4	5	6	7
$u_A(1, s_B)$	3.5	1	1.5	2	2.5	3	3.5
$u_A(2, s_B)$	6	3.5	2	2.5	3	3.5	4
difference	+2.5	+2.5	+0.5	+0.5	+0.5	+0.5	+0.5

Seven strict inequalities, all positive: $1 \prec 2$ (and $7 \prec 6$ by symmetry). For $s_B \geq 3$ the gain is always $\frac{1}{2}$: moving from 1 to 2 shifts the midpoint half a block toward B.

Is any block strictly dominant? No. A dominant block would have to be the best reply to every s_B. Against $s_B = 1$ the unique best reply is block 2 ($u_A = 6$; block 4 gives only 5), and against $s_B = 7$ it is block 6. No block is best against both, so none is dominant — the game has **no dominant-strategy equilibrium**.

4 · Worked — (c) IESDS, round by round

Round	Surviving before	Removed (each chain)	Dominator and inequalities	Knowledge needed
1	$\{1,\dots,7\}$	1 and 7	$2 \succ 1$ (table above), $6 \succ 7$; nothing else: block 2 is the best reply to $s_B = 1$, so nothing dominates 2	rationality
2	$\{2,\dots,6\}$	2 and 6	$3 \succ 2$ against $s_B \in \{2,\dots,6\}$: $5 > 3.5$, $3.5 > 2$, then $+0.5$ at $s_B = 4, 5, 6$	each knows the other is rational
3	$\{3,4,5\}$	3 and 5	$4 \succ 3$ against $s_B \in \{3,4,5\}$: $4 > 3.5$, $3.5 > 3$, $4 > 3.5$	second-order knowledge
4	$\{4\}$	—	nothing left	—

$S^\infty = \{(4,4)\}$: both chains open on the middle block and split the avenue, 3.5 customers each. Round 2 needs the round-1 removals: in the full game block 2 is the unique best reply to a rival at 1, so 3 does not dominate 2 until block 1 is gone.

> **The median wins.** Each round, a chain moves one block toward the centre because the rival is known not to be outside it. The survivor is the block with half the customers on each side — the same logic as Hotelling's and the median voter's (Lecture 2, Tadelis §5.2.5, p. 93).

5 · Worked — (d) total walking

Total distance walked, each customer to the nearest store, $D(x,y) = \sum_{j=1}^{7} \min\{|j-x|, |j-y|\}$:

$$D(4,4) = 3+2+1+0+1+2+3 = \mathbf{12}.$$

Splitting the avenue does better. With stores at 2 and 5: blocks 1–3 walk $1+0+1=2$, blocks 4–7 walk $1+0+1+2=4$, total **6**. The minimum over all pairs is 6, reached at $\{2,5\}$, $\{2,6\}$ and $\{3,6\}$: each store sits at a median of the customers it serves.

Profile	Customers walk	Chains' customers
IESDS $(4,4)$	12	3.5, 3.5
$(2,5)$	6	3, 4

IESDS doubles the walking. The chains do not care: the game is constant-sum, so *every* profile is Pareto optimal **for the chains** ($u_A + u_B = 7$ — no profile makes one better off without hurting the other). The inefficiency falls entirely on customers, who are not players.

6 · C — (e) what drives the prediction (Cabral §14.2)

Assumption	If dropped
prices fixed and equal	**the main one**: with price competition, clustering makes the products identical and invites Bertrand pricing at cost
inelastic demand (everyone buys)	far customers may not buy; spreading out gains them
two firms, one store each	a third firm breaks the clean peeling (IESDS bites much less)
discrete blocks, uniform customers	the survivor is the **median** of the customer distribution, not the geometric middle
simultaneous choice	a leader would move differently → Lecture 3

Cabral's Hotelling model (pp. 358–359): firms at the two ends of $[0, 1]$, consumers uniform, transport cost t per unit of distance, marginal cost c. The indifferent consumer solves $tx + p_1 = t(1 - x) + p_2$, so demand is $q_i = \frac{1}{2} + \frac{p_j - p_i}{2t}$ (14.3); profit maximization gives the best response $p_i = \frac{c+t+p_j}{2}$ (14.4) and, with equal costs, $\boxed{p = c + t}$ (14.5). **The margin equals the differentiation parameter** t: with $t = 0$ this is Bertrand, price at cost.

Location with prices (p. 360): moving toward the centre has a **direct effect** (more demand at given prices, positive) and a **strategic effect** (the rival cuts its price, negative).

Price competition	Prediction	Cabral's example
weak (prices set nationally)	cluster near the centre	European bank branches
intense (identical products, local prices)	locate far apart	the two ice-cream vendors on a beach

The pharmacies of slide 31 fit the first row: chains price largely at the chain level, so the direct effect dominates and they open side by side.

Links

- Prev / next: L01-07-cournot-iesds · L01-09-formulas.
- IESDS logic: L01-06-iesds. Hotelling and voting with Nash equilibrium → Lecture 2.
- Python: `line_game`, `iesds_rounds_detail` and the walking distance, reused from `Listas/lista-lecture-01-check.py` → L01-11-python §7.
- Preparation (unsolved, different streets): `Listas/prep-mini-case-lecture-01.md`.
- Practice: Tadelis 4.10 (p. 78) = `Listas/pratica-lecture-01.md` item 18; `Listas/lista-lecture-01.tex` Q5–Q6 (weighted towns, ring road, general n, continuum); quiz `Simulados/quiz-lecture01-3-iesds-2026-10-13.md` (section `space`).

Lecture 1 · Formula sheet — every definition, result and number

Prev: L01-08-spatial-competition · Next: L01-10-pitfalls · Map: L01-00-map

---

0 · The exam specification, mapped

The slides' wrap-up (slide 34) lists what the lecture covered; the course learning goals (slide 7) ask you to recognize, solve, and model. Item by item:

#	Wrap-up item (slide 34)	What you must be able to do	Note
1	Normal-form games	write $G = \langle I, \{S_i\}, \{u_i\} \rangle$ for a verbal story; fill a bimatrix; piecewise payoffs for continuous strategies	L01-01-games-normal-form
2	Static vs dynamic; complete vs incomplete information	define each; state common knowledge; count contingent-plan strategies	L01-02-information-and-strategies
3	Solution concepts (slides 21–22)	define equilibrium; judge a concept by existence, uniqueness, invariance	L01-03-solution-concepts
4	Dominant and dominated strategies	apply strict and weak definitions with the right quantifier; find all dominated strategies	L01-04-dominance §1, §5
5	Dominant-strategy equilibrium	find it or show it does not exist; prove Prop. 4.1	L01-04-dominance §2–§4
6	Pareto optimality	find the Pareto set; explain why a DSE can be Pareto dominated; the partnership band $y/n < c < y$	L01-05-pareto-social-dilemmas
7	IESDS	run it round by round with the dominator and the knowledge level of each round; state and prove Prop. 4.2	L01-06-iesds
8	IESDS in continuous games	bound best responses; Cournot intervals and their limit	L01-07-cournot-iesds

#	Wrap-up item (slide 34)	What you must be able to do	Note
9	Spatial competition	location payoffs on a line; peel by IESDS; welfare comparison; the role of prices	L01-08-spatial-competition

---

1 · Definitions and results

Item	Statement	Derived in
Normal form (slide 13; Tadelis Def. 3.3, p. 47)	$G = \langle I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I}\rangle$, $u_i : S_1 \times \cdots \times S_n \to \mathbb{R}$	L01-01-games-normal-form §2
Pure strategy (Def. 3.2, p. 46)	a deterministic plan of action; profile $s = (s_1, \dots, s_n)$	§2
Others' profile (slide 14; p. 59)	$s_{-i} \in S_{-i} = \prod_{j\neq i} S_j$, $s = (s_i, s_{-i})$	§2.1
Finite game (Def. 3.4, p. 51)	finitely many players and strategies; two players ⇒ bimatrix	§2.2
Common knowledge (Def. 3.1, p. 45; slide 16)	everyone knows E, everyone knows that everyone knows E, *ad infinitum*	L01-02-information-and-strategies §2
Strategy (slide 17)	a complete contingent plan; a actions at each of m contingencies ⇒ a^m strategies	§3
Ordinal invariance	dominance, DSE and IESDS unchanged by strictly increasing transforms of each u_i	§4.1
Solution concept, equilibrium (slide 21; p. 54)	a rule selecting reasonable profiles; an equilibrium is a selected **profile of strategies**	L01-03-solution-concepts §1
Criteria (slide 22; pp. 55–56)	existence, uniqueness, invariance (payoff changes $\leq \varepsilon$)	§2
Strict invariance	if g is the smallest strict margin, perturbations with $\varepsilon < g/2$ keep every strict inequality	§3.1
Strictly dominated (slide 23; Def. 4.1, p. 60)	$u_i(s_i, s_{-i}) < u_i(s'_i, s_{-i})$ for all s_{-i}	L01-04-dominance §1
Weakly dominated (slide 23; MWG Def. 8.B.3)	$\leq$ for all s_{-i}, $<$ for at least one	§1
Strictly (weakly) dominant (Def. 4.2, p. 61)	strictly (weakly) dominates every other strategy	§1
Claim 4.1 (p. 60) = Lesson 1	a rational player never plays a strictly dominated strategy	§1
Strict DSE (Def. 4.3, p. 61; slide 25)	every s_i^D strictly dominant	§2

Item	Statement	Derived in
Prop. 4.1 (p. 61; slide 25)	a strict DSE is the unique DSE	§2 (proof)
Weakly dominant: uniqueness	under the slides' definition a player has at most one weakly dominant strategy; under Tadelis's remark (p. 63) not	§7
Pareto (Def. 3.5, p. 57; slide 26)	s Pareto dominates s': $u_i(s) \geq u_i(s')\ \forall i$, $>$ for some i; Pareto optimal: not dominated	L01-05-pareto-social-dilemmas §1
Lesson 2 (slide 26)	individually rational choices can lead to socially undesirable outcomes	§2
IESDS (slide 29)	$S_i^0 = S_i$; $S_i^k = \{s_i \in S_i^{k-1} : \nexists s_i' \in S_i^{k-1},\ u_i(s_i', s_{-i}) > u_i(s_i, s_{-i})\ \forall s_{-i} \in S_{-i}^{k-1}\}$; $S_i^\infty = \bigcap_k S_i^k$	L01-06-iesds §2
Iterated-elimination equilibrium (Def. 4.4, p. 65)	any profile surviving IESDS	§2
Prop. 4.2 (p. 68; slide 29)	a strict DSE uniquely survives IESDS (after one round)	§2 (proof)
Order independence (MWG pp. 239–240)	strict: the end set does not depend on the order; weak: it can	§6
Round k knowledge	round 1: rationality; round k: $(k-1)$-th order knowledge of rationality	§3

1.1 · Model formulas

Model	Formula	Note
Partnership, n partners	$u_i = \frac{y \cdot \#E}{n} - c \cdot \mathbb{1}[E]$; $u(E) - u(S) = \frac{y}{n} - c$ for every s_{-i}	L01-04-dominance §4
Partnership dilemma	S dominant and all-E Pareto better $\iff \frac{y}{n} < c < y$	L01-05-pareto-social-dilemmas §3
Mafia-modified PD	M strictly dominant $\iff z > 1$	§4
Second-price auction	$u_i = v_i - \max_{j \neq i} b_j$ if i wins; bidding v_i weakly dominant	L01-04-dominance §6
Cournot, $p = a - bq$, costs c_i	$BR_i(q_j) = \max\{0, \frac{a - c_i - bq_j}{2b}\}$; $S_i^k = [BR_i(U_j^{k-1}), BR_i(L_j^{k-1})]$	L01-07-cournot-iesds §3–§5
Cournot dominance step	$v_i(x, q_j) - v_i(z, q_j) = (x - z)(a - c_i - b(x + z + q_j))$	§3
Cournot point	$q_i^* = \frac{a - 2c_i + c_j}{3b}$, $p^* = \frac{a + c_1 + c_2}{3}$, $v_i^* = \frac{(a - 2c_i + c_j)^2}{9b}$; exit if $c_i \geq \frac{a + c_j}{2}$	§5
Location, 7 blocks	$u_A = \frac{s_A + s_B - 1}{2}$ $(s_A < s_B)$, $\frac{7}{2}$ $(=)$, $7 - \frac{s_A + s_B - 1}{2}$ $(>)$	L01-08-spatial-competition §2

Model	Formula	Note
Hotelling (Cabral (14.3)–(14.5))	$q_i = \frac{1}{2} + \frac{p_j - p_i}{2t}$; $p_i = \frac{c+t+p_j}{2}$; $p = c + t$	§6

---

2 · Every number in the lecture

2.1 · Grade game (slides 10–12)

Grades: (α, α) B−/B−, (α, β) A/C, (β, β) B+/B+. Payoffs: $(0, 0)$, $(3, -1)$, $(-1, 3)$, $(1, 1)$. α strictly dominant ($0 > -1$, $3 > 1$); DSE (α, α), Pareto dominated by (β, β).

2.2 · Prisoner's dilemma (slide 24; Tadelis p. 48)

$(M, M) = (-2, -2)$, $(M, F) = (-5, -1)$, $(F, M) = (-1, -5)$, $(F, F) = (-4, -4)$. F strictly dominant (margins 1 and 1); DSE (F, F); Pareto set $\{(M, M), (M, F), (F, M)\}$. Altruistic version (p. 53): $(-3, -3)$, $(-5\frac{1}{2}, -3\frac{1}{2})$, $(-3\frac{1}{2}, -5\frac{1}{2})$, $(-6, -6)$, M dominant. Mafia: $z > 1$.

2.3 · Matrix (4.1) = slide 27 (Tadelis p. 64)

Rows U, M, D; columns L, C, R: $(4, 3), (5, 1), (6, 2)$ / $(2, 1), (8, 4), (3, 6)$ / $(3, 0), (9, 6), (2, 8)$. No dominant strategy; C $\prec$ R; IESDS: C (round 1), M and D (round 2), R (round 3) $\Rightarrow (U, L)$, payoffs $(4, 3)$, Pareto dominated by $(8, 4)$ and $(9, 6)$. Pareto set $\{(D, C), (D, R)\}$.

2.4 · Cournot (Tadelis pp. 49, 65–68)

Quadratic costs (p. 49): competitive $q_i = 25$, $p = 50$, profit 625; $q_1 = 24$ gives 648. Linear, $p = 100 - q$, $c = 10$: $BR = (90 - q_j)/2$; intervals $[0, 45]$, $[22.5, 45]$, $[22.5, 33.75]$, $[28.125, 33.75]$, …; width $45/2^{k-1}$; limit $q = 30$, $p = 40$, profit 900; at 20 each: 1000; joint optimum 22.5 each: 1012.5.

2.5 · Pharmacy avenue (slides 32–33)

$u_A(1, \cdot) = (3.5, 1, 1.5, 2, 2.5, 3, 3.5)$, $u_A(2, \cdot) = (6, 3.5, 2, 2.5, 3, 3.5, 4)$: $1 \prec 2$. No dominant block. IESDS removes $\{1, 7\}$, $\{2, 6\}$, $\{3, 5\} \Rightarrow (4, 4)$, 3.5 customers each. Walking: 12 at $(4, 4)$, minimum 6 at $\{2, 5\}, \{2, 6\}, \{3, 6\}$.

2.6 · Other book examples

Example	Numbers	Note
Advertising game (Tadelis p. 60)	L $\prec$ M ($8 > 6$, $4 > 2$, $1 > 0$); M, H undominated	L01-04-dominance §5
Battle of the Sexes (Tadelis p. 54)	$(2, 1), (0, 0)/(0, 0), (1, 2)$: nothing dominated	L01-03-solution-concepts §2
Gibbons Fig. 1.1.1 (pp. 5–7)	Right, Down, Left removed ⇒ (Up, Middle)	L01-06-iesds §4
MWG Fig. 8.B.3 (p. 238)	weak elimination ends at (D, R), (D, L) or $\{D\} \times \{L, R\}$ by order	L01-06-iesds §6
M&R Table 5.1 (p. 138)	$(-1, -1)$, $(3, -2)$, $(-2, 3)$, $(2, 2)$; grab dominant; no investment	L01-05-pareto-social-dilemmas §4

Example	Numbers	Note
Cabral Fig. 7.1 (p. 161)	$(5,5)$ vs $(4,4)$; B dominant ($6>5$, $4>3$)	§2
Partnership default (companion)	$y=6$, $c=4$, $n=2$: $(2,2),(-1,3),(3,-1),(0,0)$ = PD +4	§3

Lecture 1 · Pitfalls and a True/False self-test

Prev: L01-09-formulas · Next: L01-11-python · Map: L01-00-map

---

Part I — the confusions this lecture produces

1 · "For each s_{-i} something beats it" is not "dominated"

Dominance needs **one** alternative that wins against **every** s_{-i}. Counterexample: rows T = $(3, 0)$, M = $(1, 1)$, B = $(0, 3)$ against L, R. For each column some row beats M, yet neither T nor B beats it in both columns; M is not dominated by a pure strategy (L01-04-dominance §7.1).

2 · Strict vs weak

Weak dominance allows ties. It does not follow from rationality alone, is not robust to small payoff changes, and its iterated version depends on the order (MWG Figure 8.B.3 ends at three different places; L01-06-iesds §6). Truthful bidding in a second-price auction is weakly, **not** strictly, dominant (L01-04-dominance §6).

3 · Equilibrium = strategies, not payoffs

"The equilibrium is $(-4, -4)$" is wrong; it is (F, F), with payoffs $(-4, -4)$ (Tadelis p. 61).

4 · Pareto optimal vs equilibrium vs fair

The PD's equilibrium (F, F) is the only profile that is **not** Pareto optimal; (M, F) and (F, M) **are** Pareto optimal although grossly unequal (Tadelis p. 57). Pareto optimality says "no waste", not "best" and not "equal" (L01-05-pareto-social-dilemmas §1).

5 · Mutual knowledge is not common knowledge

"Everyone knows E" is level 1; IESDS round k needs level $k-1$ of knowledge **of rationality**. The exam announcement heard by a daydreaming student fails at level 2 (Tadelis p. 45; L01-02-information-and-strategies §2).

6 · A strategy is not an action (once anyone observes anything)

If Pair sees your letter before writing, Pair has $2^2 = 4$ strategies, not 2. Writing the matrix with 2 columns throws away the contingent plans (L01-02-information-and-strategies §3.1).

7 · "Dominant strategy" vs "the dominant-strategy outcome is good"

In the partnership, S is dominant whenever $c > y/n$; it is **also efficient** only when $c \geq y$. The dilemma is the band between (L01-05-pareto-social-dilemmas §3).

8 · IESDS in one shot vs iteratively

In matrix (4.1) M and D are **not** dominated in the original game; they become dominated only after C is removed. Checking dominance once and stopping misses rounds 2 and 3 (L01-06-iesds §3).

9 · IESDS reaches its limit, not a point, in Cournot

After any finite number of rounds the Cournot interval has width $45/2^{k-1} > 0$; the prediction $q = 30$ is the intersection of all rounds (L01-07-cournot-iesds §4). And with three firms the interval stops shrinking after round 1 (§6).

10 · The location game's survivor needs fixed prices

The middle-block prediction on the avenue is a result about locations **at given, equal prices**. With price competition, clustering invites a price war and firms may spread out (Cabral p. 360; L01-08-spatial-competition §6).

11 · "Weakly dominant equilibria need not be unique" depends on the definition

True with Tadelis's remark (p. 63: $\leq$ everywhere, no strictness required); false with the slides' definition (strict somewhere), under which each player has at most one weakly dominant strategy (L01-04-dominance §7).

Part II — True/False (original items)

Scoring as in the course: $\max\{0,\ \text{correct} - \text{wrong}/2\}$. Cover the verdict column, decide, then read the justification.

#	Claim	Verdict
1	In a static game the players must choose at the same instant.	F
2	Complete information requires that the payoff functions be common knowledge, not merely known by everyone.	T
3	If s_i is strictly dominated, then s_i is not a best response to any $s_{-i} \in S_{-i}$.	T
4	If for every s_{-i} some strategy of player i pays more than s_i, then s_i is strictly dominated.	F
5	A game can have two different strict dominant-strategy equilibria.	F
6	Replacing every payoff u_i by e^{u_i} can change which strategies survive IESDS.	F
7	In the prisoner's dilemma of the slides, (M, F) is Pareto optimal.	T
8	A strict dominant-strategy equilibrium is always Pareto optimal.	F

#	Claim	Verdict
9	In the n-partner game, effort is strictly dominant if and only if $c < y$.	F
10	Raising n at fixed y and c can turn effort from dominant into a social dilemma.	T
11	Bidding one's value in a second-price auction is a strictly dominant strategy.	F
12	In a finite game, the set that survives IESDS does not depend on the order in which strictly dominated strategies are removed.	T
13	In a finite game, IESDS can eliminate every strategy of some player.	F
14	The second round of IESDS in matrix (4.1) requires player 1 to know that player 2 is rational.	T
15	If IESDS leaves a unique profile, that profile is a strict dominant-strategy equilibrium.	F
16	In Tadelis's Cournot game, after three rounds of IESDS each firm's surviving set is $[22.5, 33.75]$.	T
17	In that game, IESDS selects $q_1 = q_2 = 30$ after finitely many rounds.	F
18	In linear Cournot with an interior solution, a higher rival cost c_2 raises firm 1's predicted quantity.	T
19	On the 7-block avenue, block 4 is a strictly dominant location.	F
20	The IESDS outcome on the avenue minimizes customers' total walking distance.	F
21	In the location game every profile is Pareto optimal for the two chains.	T
22	A mafia sanction z on finking makes (M, M) the strict dominant-strategy equilibrium for any $z > 0$.	F

Justifications

1. **F.** "Simultaneous" means without observing the others' choices; timing is irrelevant (Tadelis p. 44; Gibbons p. 4). Studying two days apart is static if neither sees the other's effort.
2. **T.** Tadelis requires the four components, payoffs included, to be common knowledge (p. 45); mutual knowledge would not support reasoning about the others' reasoning.
3. **T.** If s_i' beats s_i at every s_{-i}, then at any given s_{-i} the strategy s_i is not optimal (Tadelis Prop. 4.3, p. 70).
4. **F.** The quantifiers are in the wrong order: dominance needs one s_i' against all s_{-i}. Rows $(3,0),(1,1),(0,3)$: M is beaten in each column by a different row, and by no single row (Part I, 1).
5. **F.** Prop. 4.1: two strictly dominant strategies for the same player would each have to beat the other everywhere.
6. **F.** e^x is strictly increasing, so every inequality between two of i's payoffs at a fixed s_{-i} is preserved; dominance is ordinal (L01-02-information-and-strategies §4.1).
7. **T.** Only (M, F) gives player 2 as much as -1, so no other profile keeps player 2 at least as well off; nothing Pareto dominates it (Tadelis p. 57).
8. **F.** The PD's (F, F) is Pareto dominated by (M, M) — Lesson 2.
9. **F.** Effort is dominant iff $c < y/n$; $c < y$ is the condition for all-E to Pareto dominate all-S.
10. **T.** The threshold y/n falls with n while y does not: e.g. $y = 6$, $c = 2.5$ — effort dominant at $n = 2$ ($3 > 2.5$), dilemma at $n = 3$ ($2 < 2.5 < 6$).
11. **F.** Weakly: against a highest rival bid below both one's value and the alternative bid, the two bids pay the same.
12. **T.** MWG pp. 239–240 (Exercise 8.B.4): a strategy dominated against a set of profiles stays dominated against any subset.
13. **F.** In a finite game, a strategy that is best against some fixed surviving s_{-i} cannot be strictly dominated, so each S_i^k stays nonempty. (With infinite sets it can happen: $S = [0,1)$, $u = s$.)
14. **T.** Round 2 removes M and D because column C is gone, which player 1 can use only if he knows player 2 is rational (L01-06-iesds §3).
15. **F.** Matrix (4.1) has a unique survivor (U, L) but no dominant strategy for either player.
16. **T.** $[0,45] \to [22.5,45] \to [22.5,33.75]$ (Tadelis p. 66).
17. **F.** The width is $45/2^{k-1} > 0$ after every finite round; 30 is the limit.
18. **T.** $q_1^* = (a - 2c_1 + c_2)/(3b)$, so $\partial q_1^*/\partial c_2 = 1/(3b) > 0$: a weaker rival produces less, and firm 1's best response to less is more (L01-07-cournot-iesds §5).
19. **F.** Against a rival at block 1, block 2 pays 6 and block 4 only 5; no block is a best reply to every location.
20. **F.** $(4,4)$ gives total walking 12; $\{2,5\}$ gives 6.
21. **T.** The game is constant-sum ($u_A + u_B = 7$), so any gain for one chain is a loss for the other.
22. **F.** M beats F for each player iff $z > 1$ (Tadelis pp. 62–63); with $0 < z \leq 1$ fink stays (weakly, at $z = 1$) at least as good.

Quizzes generated from this unit belong in `Simulados/` (`/quiz-gen` can take L01-09-formulas and this note as source); the existing ones are `Simulados/quiz-lecture01-{1,2,3}-*.md`.

Links

- Map: L01-00-map · formulas: L01-09-formulas · code: L01-11-python.
- Style rules for T/F items: rules/00_estilo_avaliacao.

Lecture 1 · Computation in Python

> Every routine the lecture needs, each followed by `assert` checks against the slides' and the books' own numbers. Python only, numpy optional. The blocks run top to bottom from the repository root as one script (the unit's check does exactly that).
>
> Prev: L01-10-pitfalls · Map: L01-00-map · Formulas: L01-09-formulas

Reuse. `Listas/lista-lecture-01-check.py` already implements and tests IESDS, the Cournot interval recursion, the location game and the second-price payoff. Importing it would run its 1,500-draw sympy checks, so §0 loads **only its function definitions**. New code here covers what that file does not have: single-step dominance tests, Pareto sets, and the walking distance. `Notas/L01/figures/make_figures.py` and `companions/check_companions.py` use the same loader.

---

0 · Load the existing routines

```
import ast, itertools, pathlib, random
from fractions import Fraction as Fr
from functools import lru_cache

def lista_functions(path="Listas/lista-lecture-01-check.py"):
    """Execute only the `def` statements of the lista check script; return them as a
    ↪ dict."""
    tree = ast.parse(pathlib.Path(path).read_text(encoding="utf-8"))
    defs = [n for n in tree.body if isinstance(n, ast.FunctionDef)]
    ns = {"Fr": Fr, "itertools": itertools, "lru_cache": lru_cache, "random": random}
    exec(compile(ast.Module(body=defs, type_ignores=[]), path, "exec"), ns)
    return ns

L = lista_functions()
iesds2, rounds_detail = L["iesds2"], L["iesds_rounds_detail"]
assert {"iesds2", "cournot_iesds", "line_game", "spa_payoff", "weak_outcomes"} <=
 ↪ set(L)
```

1 · Games as payoff matrices → L01-01-games-normal-form

Two-player finite games are pairs of matrices `U1[r][c]`, `U2[r][c]` (rows = player 1).

```
PD_1 = [[-2, -5], [-1, -4]]          # rows/cols (M, F), Tadelis p. 48
PD_2 = [[-2, -1], [-5, -4]]
GRADE_1 = [[0, 3], [-1, 1]]          # rows/cols (alpha, beta), slide 12
GRADE_2 = [[0, -1], [3, 1]]
M41_1 = [[4, 5, 6], [2, 8, 3], [3, 9, 2]]   # matrix (4.1), Tadelis p. 64 = slide 27
M41_2 = [[3, 1, 2], [1, 4, 6], [0, 6, 8]]

def partnership(y, c):
    """2x2 partnership of slide 4, rows/cols (E, S): half of output minus own effort
    ↪ cost."""
    U1 = [[y - c, y / 2 - c], [y / 2, 0]]
    return U1, [list(r) for r in zip(*U1)]       # symmetric game: U2 = U1
    ↪ transposed
```

```
# Tadelis's quadratic-cost Cournot (p. 49): competitive benchmark and the deviation
 ↪  to 24
v = lambda qi, qj: (100 - qi - qj) * qi - qi**2 if qi + qj < 100 else -qi**2
assert v(25, 25) == 625 and v(24, 25) == 648
```

2 · Dominance, dominant-strategy equilibrium, Pareto → L01-04-dominance, L01-05-pareto-social-dilemmas

```
def payoff_rows(U1, U2, player):
    """Player's own payoffs as rows indexed by his strategies (columns = opponent
     ↪  strategies)."""
    return U1 if player == 0 else [list(r) for r in zip(*U2)]

def dominates(a, b, weak=False):
    """Does payoff row a dominate payoff row b (strictly, or weakly with one
     ↪  strict)?"""
    d = [x - y for x, y in zip(a, b)]
    return all(t >= 0 for t in d) and any(t > 0 for t in d) if weak else all(t > 0
     ↪  for t in d)

def dominated(U1, U2, player, weak=False):
    """Strategies of `player` dominated by some other pure strategy: {s:
     ↪  [dominators]}."""
    P = payoff_rows(U1, U2, player)
    out = {s: [t for t in range(len(P)) if t != s and dominates(P[t], P[s], weak)]
 ↪  for s in range(len(P))}
    return {s: d for s, d in out.items() if d}

def dominant(U1, U2, player, weak=False):
    """The (strictly or weakly) dominant strategy of `player`, or None."""
    P = payoff_rows(U1, U2, player)
    for s in range(len(P)):
        if all(dominates(P[s], P[t], weak) for t in range(len(P)) if t != s):
            return s
    return None

def dse(U1, U2, weak=False):
    """Dominant-strategy equilibrium (row, col) or None (Tadelis Def. 4.3)."""
    s1, s2 = dominant(U1, U2, 0, weak), dominant(U1, U2, 1, weak)
    return None if s1 is None or s2 is None else (s1, s2)

def pareto_optimal(U1, U2):
    """Profiles not Pareto dominated by any other (Tadelis Def. 3.5, p. 57)."""
    cells = {(r, c): (U1[r][c], U2[r][c]) for r in range(len(U1)) for c in
 ↪  range(len(U1[0]))}
    return {s for s, u in cells.items() if not any(dominates(w, u, weak=True) for t,
     ↪  w in cells.items() if t != s)}

assert dse(PD_1, PD_2) == (1, 1)                                    # (F, F), slide 24
```

```
assert pareto_optimal(PD_1, PD_2) == {(0, 0), (0, 1), (1, 0)}      # all but (F, F),
 ↪  p. 57
assert dse(GRADE_1, GRADE_2) == (0, 0) and (1, 1) in pareto_optimal(GRADE_1, GRADE_2)
# Example 3 (slide 27): no dominant strategy; only C is dominated (by R)
assert dse(M41_1, M41_2) is None and dominated(M41_1, M41_2, 0) == {}
assert dominated(M41_1, M41_2, 1) == {1: [2]}
assert pareto_optimal(M41_1, M41_2) == {(2, 1), (2, 2)}            # (D, C) and (D,
 ↪  R)
# Advertising game (Tadelis p. 60): L dominated by M, nothing else
AD_1 = [[6, 2, 0], [8, 4, 1], [4, 3, 2]]
AD_2 = [list(r) for r in zip(*AD_1)]
assert dominated(AD_1, AD_2, 0) == {0: [1]}
# Never best, yet not dominated by a pure strategy (note 04, section 7.1)
assert dominated([[3, 0], [1, 1], [0, 3]], 0, 0 * 3, 0) == {}
# Weak dominance: unique weakly dominant strategy under the slides' definition (note
 ↪  04, section 7)
assert dominant([[1, 1], [1, 0]], [[0, 0], [0, 0]], 0, weak=True) == 0
assert dominant([[1, 1], [1, 1]], [[0, 0], [0, 0]], 0, weak=True) is None   #
 ↪  identical rows: neither
```

3 · IESDS → L01-06-iesds

```
# Matrix (4.1): C, then M and D, then R (Tadelis pp. 64-65)
R, C, rounds = rounds_detail(M41_1, M41_2)
assert (R, C) == ([0], [0]) and rounds == [([], [1]), ([1, 2], []), ([], [2])]
# Prop. 4.2: a strict DSE is the unique survivor, after one round
assert rounds_detail(PD_1, PD_2)[:2] == ([1], [1]) and len(rounds_detail(PD_1,
 ↪  PD_2)[2]) == 1
# Gibbons Figure 1.1.1 (pp. 5-7): (Up, Middle)
G1 = [[1, 1, 0], [0, 0, 2]]; G2 = [[0, 2, 1], [3, 1, 0]]
assert iesds2(G1, G2)[:2] == ([0], [1])
# No bite: Battle of the Sexes (Tadelis p. 54) and Gibbons Figure 1.1.4 (p. 7)
assert iesds2([[2, 0], [0, 1]], [[1, 0], [0, 2]])[:2] == ([0, 1], [0, 1])
F114_1 = [[0, 4, 5], [4, 0, 5], [3, 3, 6]]; F114_2 = [[4, 0, 3], [0, 4, 3], [5, 5,
 ↪  6]]
assert iesds2(F114_1, F114_2)[:2] == ([0, 1, 2], [0, 1, 2])
# Weak elimination depends on the order: MWG Figure 8.B.3 (p. 238)
W1 = [[5, 4], [6, 3], [6, 4]]; W2 = [[1, 0], [0, 1], [4, 4]]
def weak_orders(U1, U2, R=None, C=None):
    """End states of every one-at-a-time order of weak elimination (the lista's
    weak_outcomes is hard-wired to 3x3, so this generalizes it to any size)."""
    R = tuple(range(len(U1))) if R is None else R
    C = tuple(range(len(U1[0]))) if C is None else C
    moves = [("r", r) for r in R if any(dominates([U1[t][c] for c in C], [U1[r][c]
 ↪  for c in C], True) for t in R if t != r)]
    moves += [("c", c) for c in C if any(dominates([U2[r][t] for r in R], [U2[r][c]
 ↪  for r in R], True) for t in C if t != c)]
    if not moves:
        return {(R, C)}
```

```
    return set().union(*(weak_orders(U1, U2, tuple(x for x in R if x != v), C) if k
    ↪ == "r"
                         else weak_orders(U1, U2, R, tuple(x for x in C if x != v))
                         ↪ for k, v in moves))
assert weak_orders(W1, W2) == {((2,), (1,)), ((2,), (0,)), ((2,), (0, 1))}   # (D,
↪ R), (D, L), or D x {L, R}
assert iesds2(W1, W2)[:2] == ([0, 1, 2], [0, 1])                  # strict IESDS
↪ removes nothing
```

4 · Partnership and changing the game → L01-05-pareto-social-dilemmas

```
def partnership_verdict(y, c, n):
    """Dominance and dilemma verdict for n partners: gain from effort y/n - c, all-E
    ↪ vs all-S y - c."""
    g = y / n - c
    dom = "E" if g > 0 else ("S" if g < 0 else "none")
    return dom, (dom == "S" and y - c > 0)          # (dominant strategy, social
    ↪ dilemma?)

assert partnership_verdict(6, 4, 2) == ("S", True)          # default: the dilemma
assert partnership_verdict(6, 2, 2) == ("E", False)
assert partnership_verdict(6, 7, 2) == ("S", False)         # shirking dominant AND
↪ efficient
assert partnership_verdict(6, 3, 2) == ("none", False)      # c = y/n: no dominance
↪ at all
# y = 6, c = 4, n = 2 is Tadelis's PD plus 4 on every payoff
U1, U2 = partnership(6, 4)
assert [[x - 4 for x in row] for row in U1] == PD_1      # rows (E, S) <-> (M, F)
# reuse: the lista's n-partner condition with no synergy (theta = 0) is exactly y/n <
↪ c < y
assert L["partnership_conditions"](2, 6, 0, 4) and not L["partnership_conditions"](2,
↪ 6, 0, 7)

# Mafia-modified PD (Tadelis pp. 62-63): M strictly dominant iff z > 1
def mafia(z):
    return [[-2, -5], [-1 - z, -4 - z]], [[-2, -1 - z], [-5, -4 - z]]
assert dse(*mafia(1.01)) == (0, 0) and dse(*mafia(0.99)) == (1, 1) and dse(*mafia(1))
↪ is None
# M&R grabbing (Table 5.1, p. 138), A as rows (Grab, Don't)
GR_A = [[-1, 3], [-2, 2]]; GR_B = [[-1, -2], [3, 2]]
assert dse(GR_A, GR_B) == (0, 0) and (1, 1) in pareto_optimal(GR_A, GR_B)
```

The partnership's rows are (E, S) and the PD's (M, F): effort plays the role of mum (cooperate), shirking the role of fink (defect).

5 · Cournot by IESDS → L01-07-cournot-iesds

```
cournot_iesds = L["cournot_iesds"]      # [(lows, highs) per round], exact Fractions

hist = cournot_iesds(Fr(100), Fr(1), Fr(10), Fr(10), 12)
```

```
assert [(h[0][0], h[1][0]) for h in hist[:4]] == [(0, 45), (Fr(45, 2), 45), (Fr(45,
 ↪  2), Fr(135, 4)), (Fr(225, 8), Fr(135, 4))]
assert all(h[1][0] - h[0][0] == Fr(45, 2 ** k) for k, h in enumerate(hist))      #
 ↪  width 45 / 2^(k-1)
assert hist[-1][0][0] < 30 < hist[-1][1][0]                                       #
 ↪  never a point in finite rounds

def cournot_eq(a, b, c1, c2):
    """Interior Cournot point q_i = (a - 2c_i + c_j) / (3b), price and profits."""
    q1, q2 = (a - 2 * c1 + c2) / (3 * b), (a - 2 * c2 + c1) / (3 * b)
    p = a - b * (q1 + q2)
    return q1, q2, p, (p - c1) * q1, (p - c2) * q2

assert cournot_eq(100, 1, 10, 10) == (30, 30, 40, 900, 900)                       #
 ↪  Tadelis p. 67
profit = lambda q, qj: (90 - q - qj) * q
assert profit(20, 20) == 1000 and profit(22.5, 22.5) == 1012.5                    # p.
 ↪  68; joint optimum
# round-1 dominance identity v(45,q2) - v(q1,q2) = (q1-45)(q1-45+q2), on a grid
assert all(profit(45, q2) - profit(q1, q2) == (q1 - 45) * (q1 - 45 + q2) for q1 in
 ↪  range(0, 120) for q2 in range(0, 60))
# exit: c1 >= (a + c2)/2 drives firm 1 to zero
h = cournot_iesds(Fr(100), Fr(1), Fr(60), Fr(10), 40)
assert h[-1][1][0] == 0 and h[-1][0][1] == h[-1][1][1] == 45                      #
 ↪  firm 2 at monopoly (100-10)/2
```

6 · Second-price auction: weak, not strict → L01-04-dominance §6

```
spa = L["spa_payoff"]                       # spa(i, bids, values, reserve), exact
 ↪  Fractions
vals, grid = [Fr(5), Fr(3)], [Fr(k, 2) for k in range(0, 21)]
def pay(b):                                 # bidder 0's payoffs against every rival bid
 ↪  on the grid
    return [spa(0, (b, m), vals, Fr(0)) for m in grid + [Fr(12)]]
truth = pay(vals[0])
for b in grid:
    if b != vals[0]:
        assert dominates(truth, pay(b), weak=True)        # truthful weakly dominates
         ↪  b ...
        assert not dominates(truth, pay(b))               # ... but never strictly
```

7 · The pharmacy avenue → L01-08-spatial-competition

```
U_A, U_B = L["line_game"](7, [1] * 7)
uA = lambda a, b: Fr(a + b - 1, 2) if a < b else Fr(7, 2) if a == b else 7 - Fr(a + b
 ↪  - 1, 2)
assert all(U_A[a - 1][b - 1] == uA(a, b) for a in range(1, 8) for b in range(1, 8))
assert all(U_A[1][b] > U_A[0][b] for b in range(7))                  # (b) block 1 <
 ↪  block 2
```

```
assert dominant(U_A, U_B, 0) is None                                    # (b) no
 ↪  dominant block
R, C, rounds = rounds_detail(U_A, U_B)
assert R == C == [3] and [r[0] for r in rounds] == [[0, 6], [1, 5], [2, 4]]   # (c)
 ↪  (4, 4) in 3 rounds
D = lambda x, y: sum(min(abs(j - x), abs(j - y)) for j in range(1, 8))
best = min(D(x, y) for x in range(1, 8) for y in range(1, 8))
assert D(4, 4) == 12 and best == 6                                      # (d) IESDS
 ↪  doubles the walking
assert {(x, y) for x in range(1, 8) for y in range(x, 8) if D(x, y) == 6} == {(2, 5),
 ↪  (2, 6), (3, 6)}
print("L01 python note: all checks pass")
```

Links

- The routines' source and their randomized tests: `Listas/lista-lecture-01-check.py` (run it with `python Listas/lista-lecture-01-check.py`; needs sympy).
- Figures built from the same functions: `Notas/L01/figures/make_figures.py`. Companion checks: `Notas/L01/companions/check_companions.py`.
- Rules pack: rules/game-theory rule 12 (computational solution, verified by checking deviations); rules/lang-python.